

A Primer on Residential Mortgage Credit

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Sharad Chaudhary

646.855.9793

sharad.chaudhary@bankofamerica.com

RMBS Trading Desk Strategy

Ohmsatya Ravi

646.855.9150

ohmsatya.p.ravi@bankofamerica.com

Qumber Hassan

646.855.4308

qumber.hassan@bankofamerica.com

Vipul Jain

646.855.9109

vipul.p.jain@bankofamerica.com

Ankur Mehta

646.855.1190

ankur.mehta@bankofamerica.com

RMBS Trading Desk Modeling

ChunNip Lee

646.855.5040

chunnip.lee@bankofamerica.com

Marat Rvachev

646.855.2632

marat.rvachev@bankofamerica.com

- ▶ The term mortgage credit is shorthand for residential mortgage credit risk – the risk that a homeowner will default on their mortgage and the lender of mortgage funds will not be able to fully recover their principal. The heart of mortgage credit analysis thus revolves around understanding what causes mortgage default and how much of the loaned principal one could hope to recover if default were to happen. The goal of this primer is to provide the foundational blocks for such analysis.
- ▶ We begin with a discussion of mortgage delinquency, a crucial stepping stone on the path to default. We describe commonly used metrics for tracking the degree of mortgage delinquency in a pool of mortgages and discuss the various causes of delinquency.
- ▶ The progression from delinquency to foreclosure is not pre-ordained and, in practice, the mortgage servicer will do their utmost to avoid the outcome of foreclosure since the associated costs (economic and non-economic) can be substantial. Thus, the section on loss mitigation describes the various alternatives to foreclosure that servicers pursue in order to get the borrower back on track, or failing that, to minimize the losses they incur.
- ▶ When loss mitigation activities are ineffective or the borrower is unwilling to work with the servicer, foreclosure becomes inevitable. We explain the distinction between mortgage default and mortgage foreclosure and walk through some of the intricacies of the foreclosure process. The institutional details of the process are supplemented with a broad overview of the economic determinants of mortgage default.
- ▶ Understanding the economic determinants of delinquency and default constitute the foundation for putting together a model of default. We describe the mathematical structure of a typical model and describe how it can be used to forecast voluntary prepayments and defaults.
- ▶ The main body of our primer concludes with addressing the crucial issue of determining what fraction of the principal we should expect to lose on a defaulted loan (loss severity). We describe the components of loss severity and illustrate how it varies as a function of the economic environment and the characteristics of the mortgage loan.
- ▶ There are a number of appendices to the primer which take a deep-dive into issues that cannot be easily covered in the main body of the primer without a break in continuity. The appendices review the conventions for delinquency calculations, the causes of seasonal trends in delinquency, monitoring the impact of mortgage servicing on credit performance, state-level variation in foreclosure law, the significance of junior mortgages in foreclosure proceedings, personal bankruptcy, and the taxation of mortgage defaults.

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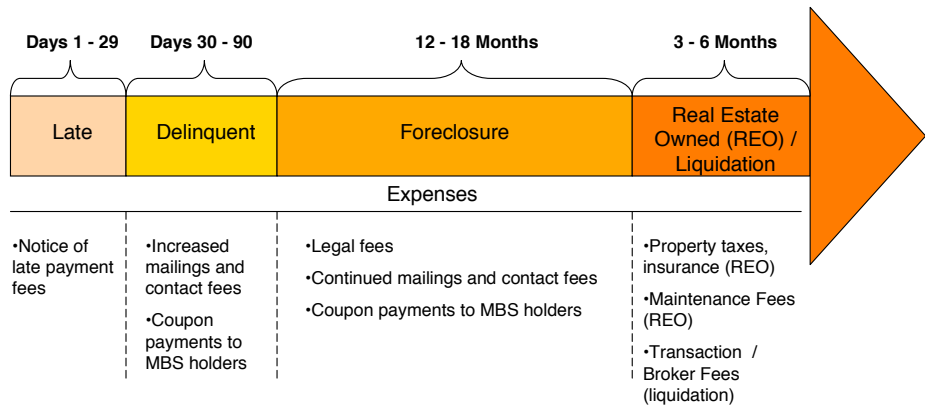
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I. INTRODUCTION

The term **mortgage credit** is shorthand for residential mortgage credit risk – the risk that a homeowner will default on their mortgage and the secured creditor (the lender of the mortgage funds) will not fully recover their principal. The heart of mortgage credit analysis thus revolves around understanding what causes mortgage default and how much of the loaned principal one could hope to recover if default were to happen. As Figure 1.1 suggests and our discussion in the following sections will confirm, mortgage default is best viewed as a multi-stage process that can last up to one to two years (or longer in extreme cases), rather than as a single discrete event. It then follows that assessing mortgage credit risk requires a detailed understanding of the entire pipeline of events depicted in the figure and the goal of this primer is to provide the foundational blocks for this understanding. While there is probably a library’s worth of material on various aspects of mortgage credit, there are very few front-to-back discussions and our primer hopes to fill this gap.

Figure 1.1: Overview of Mortgage Delinquency and Foreclosure*



*The expenses shown in the bottom half of the figure are those commonly incurred by servicers and/or mortgage investors in the delinquency and foreclosure process.
Source: MBA, Banc of America Securities

The primer is organized as follows. First, we begin with a discussion of mortgage delinquency (Section II), a crucial stepping stone on the path to default. The objectives here are describing how to calculate the level or degree of delinquency in a pool of mortgages and discuss the various causes of delinquency. The progression from delinquency to foreclosure is not pre-ordained and, in practice, the mortgage servicer will do their utmost to avoid this outcome since the costs (economic and non-economic) associated with a mortgage foreclosure can be substantial. Thus, Section III (Loss Mitigation) describes the various alternatives to foreclosure that servicers pursue in order to get the borrower back on track, or failing this objective, to minimize the amount of losses they incur. Section IV (Anatomy of Mortgage Defaults) walks us through what happens when loss mitigation activities fail and foreclosure becomes necessary. Here, we clarify the distinction between mortgage default and mortgage foreclosure and walk through some of the intricacies of the foreclosure process. These details are supplemented with a broad overview of the economic determinants of mortgage default. Section V (Modeling the Delinquency and Default Process) takes a step back from the

institutional details of the previous sections and describes how these details are captured in a mathematical model for mortgage delinquency and default. The section describes how loan-level models for forecasting voluntary prepayments and defaults work and how they are different from the typical models used to forecast prepayments on Agency MBS pools. The main body of our primer concludes with Section VI (Calculating and Modeling Loss Severity), which addresses the crucial issue of determining what fraction of the principal we expect to lose on a defaulted loan (loss severity). We describe the components of loss severity and illustrate how it varies as a function of the economic environment and the characteristics of the mortgage loan.

There are also a number of appendices to the primer which take a deep-dive into issues that cannot be easily covered in the main body of discussion without a break in continuity. The appendices review the conventions for delinquency calculations (Appendix A), the causes of seasonal trends in delinquency (Appendix B), monitoring the impact of mortgage servicing on credit performance (Appendix C), state-level variation in foreclosure law (Appendix D), the significance of junior mortgages in foreclosure proceedings (Appendix E), personal bankruptcy (Appendix F), and the taxation of mortgage defaults (Appendix G).

As should be clear even from our brief discussion of the contents of the primer, it is almost impossible to do justice to the depth and breadth of this topic without resorting to the encyclopedia format. From the perspective of depth, apart from including more detail in the appendices, we have also footnoted several papers throughout the primer that can be used for a closer look at a particular issue. From the breadth perspective, there are a couple of important issues that we do not do justice to in the body of the primer and that we briefly discuss here.

One crucial omission from our primer is **moral hazard**. While not mentioned explicitly, this is a theme that is present in any discussion of mortgage credit. At the heart of the matter is balancing the potentially conflicting goals of wanting to help homeowners who are genuinely beset with unexpected financial hardship by reducing their mortgage payments versus setting up a system that incents mortgage holders to take on more indebtedness than they should because they expect to get “bailed out” (i.e., have some amount of their debt forgiven) if their payments become unmanageable. Stated simply, debt forgiveness always has a level of moral hazard associated with it. Further complicating the issue is the fact that the obligations associated with a mortgage can be quite complex and borrowers may not always completely understand the risks associated with their loans (for example, unexpected payment increases because of a higher than expected rate reset and a subsequent lack of alternative refinancing opportunities, balance increases for negative amortization loans etc.).

A second omission is **mortgage fraud** – a borrower’s misrepresentation of their willingness and/or ability to repay the loan. This risk is non-trivial in most structured finance transactions because most investors buy a fractional share of a large pool of mortgages and consequently do not have the wherewithal to perform the necessary due-diligence on every loan. As a way to manage this risk, investors depend on a number of third parties associated with the origination and securitization process. Briefly, these third parties have the following roles and functions:

- **Originator:** Lends money to the homeowner and underwrites the mortgage loan to a pre-specified set of standards;
- **Seller:** Makes “reps and warranties”¹ on the mortgage loans being pooled into the MBS;
- **Servicer:** (1) Collects and forwards borrower payments; (2) Advances payments on delinquent loans or loans in foreclosure; (3) Follows workout or foreclosure procedures with delinquent borrowers.
- **Underwriter:** Structures the MBS and places the resulting bonds with institutional investors;
- **Trustee:** Custodian, registrar and paying agent for the Trust. Acts as a fiduciary to protect the interests of MBS holders.
- **PMI provider:** Underwriter of insurance policies for loans with loan-to-value ratios above 80%.

In general then, the MBS investor is exposed to **third-party risk**, the risk that any of the entities listed above do not do their jobs adequately and therefore leave the investor exposed to mortgage credit risk. For example, the insolvency of the seller dissolves the reps and warranties associated with the PSA which in turn means that the investor has no recourse in the case of mortgage fraud. This primer does offer a fairly extensive discussion of mortgage servicers since they play an integral part in managing the borrower’s path through delinquency and as such can have a dramatic influence on credit performance.²

¹ Representations and warranties refer to the seller’s guarantee on the accuracy of the information and underwriting guidelines for the mortgage loans backing the transaction. The guarantee is backed by the seller’s promise to repurchase (or cure/substitute) any loans that are not consistent with its representations. The Pooling and Servicing Agreement (PSA) is the legal document that specifically codifies the reps and warranties for a specific structured finance transaction.

² See Section III and Appendix C for further details.

II. MORTGAGE DELINQUENCY

Introduction

As Figure 1.1 showed, the onset of mortgage delinquency is a crucial stepping stone on the path to default and thus serves as a natural starting point for a discussion of mortgage credit. Delinquency statistics are probably the most commonly used tool to monitor the credit health of a pool of mortgages and we begin by profiling some of the commonly used delinquency statistics and describing how they are constructed and used in practice. A separate objective is to also discuss some of the common causes of mortgage delinquency and the importance of monitoring transitions through different degrees of delinquency.

Defining and Measuring Delinquency

The mortgage borrower's monthly payment is due on the 1st of every month but, given that most mortgage contracts allow a two week grace period, the mortgage delinquency clock only starts ticking once a servicer fails to receive the borrower's payment by the 16th of the month. The time elapsed on this clock is usually reported in increments of months based on either the MBA or OTS industry standard. The details of these two conventions are described in Appendix A but the basic point behind the more commonly used MBA convention is as follows. If the borrower's payment, which is due on the first of the month, is not collected by the **cut-off date** (typically the end of the same month), then the borrower will be reported as being 30-days late. If this payment is not made up by the next cut-off date (the end of the next month), the borrower is reported as 60-days late and so on. The different degrees of delinquency are usually referred to as **early-stage delinquency** (30-59 days delinquent), **late-state delinquency** (90+ days delinquent), and **severely delinquent** (same as late-stage delinquency). One of the critical points on the delinquency clock is when the borrower is 3 monthly payments behind (the onset of **severe delinquency**) since this is the juncture at which state property laws allow servicers to commence the foreclosure process. It is possible for the borrower's delinquency clock to stop at a 30-day delinquency (or, less frequently, a 60-day delinquency) through the process of **rolling delinquency** – the borrower misses one monthly payment and then starts making monthly payments again without making up the missed payment. This borrower would essentially keep being reported as being 30-days delinquent.

Causes of Mortgage Delinquency

Leaving measurement issues aside for the moment, we now turn to some of the causes of mortgage delinquency. In general, accidental delinquencies are fairly commonplace as it is not unusual for a borrower to occasionally send in a cheque that is a couple of days late. More serious cases of delinquency can be classified into one of three general categories:³

³ For further details, see Charles A. Capone, *Providing Alternatives to Mortgage Foreclosure: A Report to Congress*, Department of Housing and Urban Development, Washington, D.C., 1996.

- **Non-recurring Delinquencies due to Economic and Non-Economic Stress**
 - Financial stresses that can cause delinquencies include unemployment, a major expense shock (house repairs, medical expenses, legal expenses), and being over-extended with debt.
 - Non-financial stresses such as divorce or death can also lead to missed payments.
- **Recurring Delinquencies**
 - Seasonal workers (in the construction industry, for example).⁴
 - Over-extension around year-end holidays.
 - Delinquencies from borrowers who can have significant volatility in their income from month-to-month (such as subprime borrowers).
- **“Non-hardship” Delinquencies**
 - Borrowers who walk away from houses that are “underwater” (the value of the home is less than the amount of outstanding mortgage debt).

To provide a feel for the relative prevalence of the different reasons for delinquency, Figure 2.1 presents some results from a delinquency survey conducted by Freddie Mac. Taken at face value, the survey suggests that an interruption in borrower income is the single largest determinant of delinquency. However, there is probably significant selection bias in this survey since the respondents correspond to borrowers who the mortgage servicer was successfully able to contact. In other words, the delinquent borrowers whom servicers were not able to contact probably included a disproportionate number of people who walked away from their houses because they had negative equity. The other takeaway is that it is not particularly easy to build predictive models for mortgage delinquency since many of the major reasons for delinquency: loss of income, death, illness, marital difficulties etc., are hard to forecast at the borrower level.

The different causes of delinquency are also important because they have different **cure rates**. A cure rate simply captures the probability that a borrower will transition from delinquency to being current on their mortgage. A **fail rate** is the complement of this probability and captures the likelihood that the borrower will transition from delinquency to default. Revisiting Figure 2.1, we can see that several of the causes of delinquency such as “Marital difficulties”, “Property problem or casualty loss”, or “Inability to sell or rent property” have a fairly low frequency of occurrence, but can have some of the highest fail rates associated with them.⁵ Thus, even an after-the-fact collection of the reason for mortgage delinquency can be useful to the servicer in terms of predicting the conditional probability of foreclosure.

⁴ A more thorough discussion of the seasonal patterns in mortgage delinquency can be found in Appendix B.

⁵ The concept of “modified loans” that appears as a column heading in Figure 2.1 will be discussed in more detail in Section III.

Figure 2.1: Distribution of Hardship and Default Reasons for Delinquent Borrowers*

Mortgage Hardship Reason	Share of Delinquent Loans Citing Reason ¹	Fail Rate Among Modified Loans ²
Loss of income	41.8%	18.5%
<i>Unemployment</i>	<i>17.4%</i>	<i>19.7%</i>
<i>Curtailment of Income</i>	<i>22.0%</i>	<i>17.9%</i>
<i>Business Failure</i>	<i>2.3%</i>	<i>15.6%</i>
Death or Illness in the Family	23.2%	17.6%
Extreme Financial Stress Other Than Loss of Income	14.4%	14.4%
<i>Excessive Obligation</i>	<i>11.5%</i>	<i>15.0%</i>
<i>Extreme Hardship</i>	<i>2.5%</i>	<i>12.7%</i>
<i>Payment Adjustment</i>	<i>0.4%</i>	<i>14.3%</i>
Marital difficulties	7.6%	21.8%
Property problem or casualty loss	1.9%	4.7%
Inability to sell or rent property	1.3%	23.7%
Employment transfer or military service	0.8%	21.1%
All other reasons ³	9.0%	9.5%

*Based on calculations on a sample of Freddie Mac loans that went delinquent between 2001 and 2006.

¹ These loans corresponded to loans in which the borrower was successfully contacted by the servicer and underwritten for a workout.

² Among loans that were modified after being re-underwritten, the incidence of home loss through foreclosure sale, deed-in-lieu transfer, or chargeoff.

³ The category of “All other reasons” includes property abandonment, environment/energy costs, incarceration, payment disputes, fraud, servicing problems, borrower non-contact, and simply “other” reasons.

Source: Freddie Mac

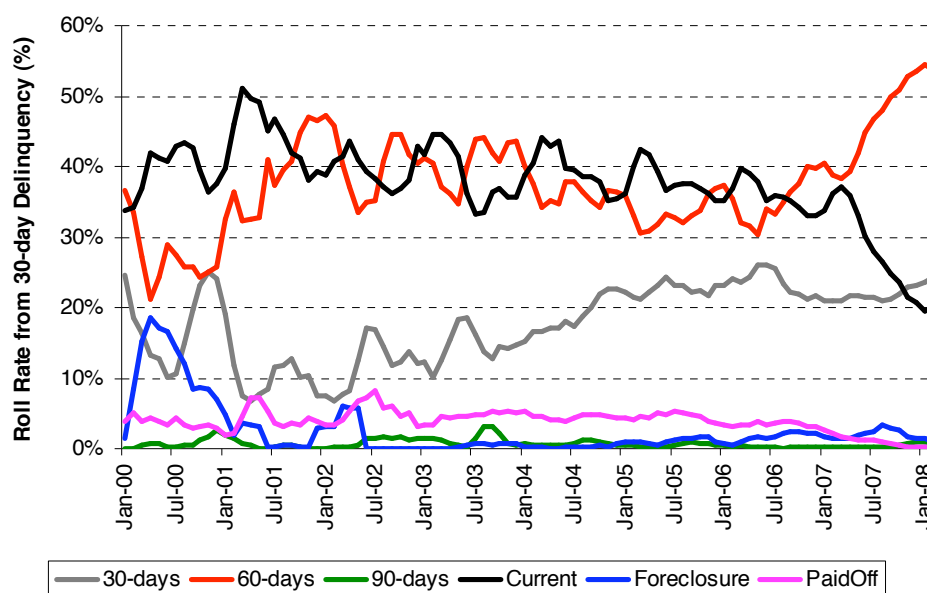
Delinquency Transitions

As our discussion on fail and cure rates suggests, apart from understanding its fundamental determinants, another important aspect of studying delinquencies is calculating the rate (“roll-rate”) at which borrowers transition from one delinquency status to another. Some of the multiple ways this roll rate data can be used includes the analysis of loss mitigation options (Section III), the measurement of servicer performance (Appendix C), and predicting the probability of default (or curing) (Section V). For example, while mortgage servicers have little ability to influence a borrower’s transition from being current on their mortgage payments to being delinquent, they do have considerably more leeway over what happens to this borrower’s delinquency status after the onset of delinquency. A servicer with excellent loss mitigation abilities may be able to coerce a high fraction of borrowers from an early stage delinquency status to being current. Similarly, the transition from early stage delinquency status to foreclosure in servicer portfolios may give us another view into a servicer’s approach to loss mitigation and in particular to how aggressively they pursue foreclosure.

To visualize delinquency transitions, take a look at Figure 2.2, which graphs the transition probability from the 30-day delinquent status to other statuses (listed at the bottom of the figure) for the portfolio of a large subprime servicer. Notice that over the time period covered

by the graph, the overwhelming majority of transitions from the 30-day delinquent state were to 30-days (a rolling delinquency), 60-days (falling even further behind), or back to current (“cures”).⁶ These individual transition probabilities clearly vary as a function of the economic environment and one can easily see the impact of the 2007/2008 housing crisis in the drop in cure rates (transitions from “30-days” to “Current”) that shows itself from the first quarter of 2007. The corresponding increase in the transition rate from 30-days to 60-days delinquency suggests that most of this increase can potentially be explained by relatively more borrowers moving deeper into delinquency than before.

Figure 2.2: Delinquency Transitions for Subprime Loans from 30-days Delinquent to Other States*



*For loans with ages between 12 to 23 months.

Source: Banc of America Securities, Loan Performance

Summarizing our discussion, apart from measuring the overall proportion of borrowers in the different stages of delinquency, it is also useful to track the rate at which these borrowers move from one of these stages to another. This rate can be heavily influenced by the actions that servicers take to cure delinquency, thereby reinforcing the idea that the path from delinquency to foreclosure is not a simple straight line, but can include several back-and-forth oscillations from one state to another. The servicer-delinquent borrower interaction falls under the general topic of loss mitigation and is what we will focus on in the next section. A deeper look at the impact of servicers on credit performance is presented in Appendix C.

⁶ The non-zero 30-day to 90-day transitions in the figure are probably an artifact of bad reporting.

III. LOSS MITIGATION

Introduction

Once a borrower falls behind on their mortgage payments, the mortgage servicer will customarily attempt to search for cost-effective alternatives to the very expensive option of foreclosing the borrower's mortgage. This activity of pursuing alternatives to foreclosure by "working" with the borrower is often generally referred to as **loss mitigation** (or **workout**). These workout or loss mitigation strategies can take a number of different forms. Generally speaking, the preferred outcome of loss mitigation is to keep the borrower in their house by getting them back on track since this is the least expensive alternative for all involved parties. If the borrower is unable to keep up with the payments even after the monthly payment has been reduced, more aggressive mitigation techniques are called for and the servicer will then try to work in conjunction with the borrower to sell the house and minimize the loss incurred on the amount they are owed. Behind this high-level description are a wealth of institutional practices and details since loss mitigation is one of the most important functions of a mortgage servicer, especially when they service non-prime mortgages.

An Overview of Loss Mitigation

Much of the groundwork for a successful resolution of a borrower's delinquency is laid at the very initiation of the mortgage contract and the practices followed by the mortgage servicer subsequent to this initiation. In fact, two of the most important tasks a servicer executes are **collection management** and **default management**. Collection management involves preventing current loans from becoming delinquent and reverting early/moderate stage delinquencies to a current status. Successful collection techniques involve a judicious blend of conventional wisdom and cutting edge technology. As soon as a loan is boarded on the servicing system, the customer service department does a welcome call typically within the first 30 days prior to first payment due date. This serves the twin purposes of welcoming the customer and verifying contact information. If no contact is made, the exception is referred to the skip trace unit whose sole purpose is to track down customers. The response of subprime mortgage servicers to early stage delinquencies (30-60 days) is typically more aggressive than that of prime servicers. Intelligent automated systems are used to analyze prior borrower payment behavior and prioritize collection efforts. Scoring tools such as Freddie Mac's EarlyIndicator™ enable servicers to target high-risk loans much earlier in the delinquency process. Notices of late fees are sent and followed by repeated calling campaigns until contact is made with the borrower. Call-center technology innovations including predictive dialing systems to dramatically improve the efficacy of calling campaigns. Call-center supervisors monitor a number of key metrics to ensure efficient collections. Measures include monthly call volume, call abandon rates, average speed to answer, average talk time, and automated voice response use percentage.

If such contact attempts fail, letters demanding payment and threatening foreclosure are often sent and property inspections may be ordered. If the loan progresses to moderate stage delinquency (60 days), the collection effort is typically assumed by more experienced collectors. Contact attempts will be made on an almost daily basis and every effort is made to get the mortgage cash flowing. If all collection management efforts fail and the loan progresses to late stage delinquency (90+ days), the loan is then referred to default management. Default management involves minimizing losses on seriously delinquent contracts. The primary

philosophy of default management is to help borrowers retain possession of their property or sell it (based on the borrower's desire) and cure the loan default using appropriate resolution techniques. Relying too aggressively on foreclosure techniques will lead to higher collateral losses as a foreclosure-related home sale often results in high loss severities. Therefore, servicers typically employ a dual strategy for late stage delinquencies: foreclosure proceedings and loss mitigation. While foreclosure is usually initiated when loans become seriously delinquent, loss mitigation techniques are simultaneously attempted in order to avoid costly liquidations. Once a request for foreclosure is initiated, the servicer orders a **broker price opinion (BPO)**⁷ or an **Automated Valuation Model (AVM)**⁸ within the next 20 to 30 days. A good handle on the value of the property is necessary for the servicer to follow the best loss mitigation strategy. In some cases a full property appraisal may be required. In addition, the servicer will refresh the valuation of properties in foreclosure on a periodic basis, typically every six months.

Motivation for Loss Mitigation: The High Cost of Foreclosure

As part of our discussion above suggests, the crucial role of loss mitigation arises from the fact that foreclosure can be a very expensive outcome for the lender. This impression is actually incomplete because the borrower and also seemingly unrelated parties to the mortgage transaction also pay a high price when a loan forecloses. For example, a foreclosure sale can depress the prices of other houses in the neighborhood and can also result in substantial amounts of foregone income for the relevant municipality. More explicitly, we can itemize some of the costs incurred by the various participants in the foreclosure process as follows:⁹

Mortgage Borrower Costs:

- Loss of housing asset, amount of downpayment, principal paid;
- Penalties and finance charges levied by the servicer over the delinquency period;
- Legal fees associated with the foreclosure process;
- Taxes on the amount of forgiven debt;¹⁰
- Limited and more expensive access to credit in the future;
- Moving expenses; and,
- Psychological effects of default.

Mortgage Lender/Servicer Costs:

- Loss of principal equal to the difference between the outstanding mortgage debt and the foreclosure sale price;

⁷ A BPO is often used in lieu of a formal appraisal to provide an estimate of the likely sale price of a property. The estimate is based on collecting a set of relevant information that would influence the property price including physical characteristics of the property (square footage, number of units, basement, lot size etc.), assessment of the property condition, comparable sales, comparable listings and neighborhood data (trends in housing supply, new construction, rental market etc.). The pricing opinion is based on an exterior "drive-by" inspection and no internal inspection is performed. The final estimate includes a range of values for the sale price based on 90-120 days marketing time including an "as-is" value and an "as-repaired" value. The opinion includes photographs of the property (and comparable listings) and a description and estimate of the necessary repairs.

⁸ An AVM is a computer-generated appraisal that relies on a database of comparable sales to estimate a property sales price.

⁹ Agpar et al, *The Municipal Cost of Foreclosures: A Chicago Case Study*, Homeownership Preservation Foundation Housing Finance Policy Research Paper Number 2005-1, February 27, 2005.

¹⁰ This tax has been temporarily suspended. See Appendix G for details.

- Increased servicing expenses;
- Opportunity cost of funds tied up in foreclosure process;
- Legal fees associated with the foreclosure process;
- Potential portfolio contagion for properties from the same area as the foreclosure sale because of the “neighborhood” effect of home prices;
- Limited and more expensive access to capital markets financing; and,
- Reputational risk.

Neighborhood Costs:

- Negative impact on local property values;¹¹
- Vacant properties are associated with vandalism, drug dealing etc; and,
- Negative impact on local businesses.

Municipality Costs:

- Loss of property taxes;
- Legal expenses;
- Reduced attractiveness of neighborhood translates into slower growth in the property tax base;
- Increased policing and burden on fire departments;
- Inspection expenses and demolition costs (if necessary); and,
- Reputational damage.

The point behind this exhaustive listing is to emphasize the significant monetary and non-monetary expenses associated with a foreclosure and its spillover impact on the neighborhood and municipality. Clearly, some of these components are hard to quantify, but various studies have estimated that the explicit (monetary) costs associated with a single foreclosure can rack up as much as \$78,000 in expenses for the various participants listed above.¹² In contrast, and as suggested by the nomenclature, loss mitigation is a much more inexpensive alternative (in general).

The Building Block of Loss Mitigation: Reaching the Borrower

Despite the compelling economic benefits offered by loss mitigation, the commonsensical approach of reaching out to the borrower, understanding why they are delinquent, and trying to customize a remediation plan for them is complicated by one of the fundamental and long-standing problems of loss mitigation: delinquent borrowers are hard to reach. In practice, the contact rate of servicers and delinquent borrowers ranges from 50% to 75% according to industry participants. In other words, there are periods of time during which servicers are in communication with only half of all borrowers who are delinquent. Absent borrower contact,

¹¹ For example, a recent study estimated that after controlling for property characteristics, a single-family foreclosure lowered the value of homes located within an eighth of a mile by an average of 0.9%. See Dan Immergluck and Geoff Smith, *The External Costs of Foreclosure: The Impact of Single-family Mortgage Foreclosures on Property Values*, Housing Policy Debate, Vol. 17, Issue 1, 2006.

¹² See U.S. Congress, Senate Joint Economic Committee, *Sheltering Neighborhoods from the Subprime Foreclosure Storm*, Washington, GPO 2007.

the servicer is forced to abandon loss mitigation and commence the foreclosure process.

Why are delinquent borrowers hard to reach? Essentially, the reasons boil down to fear, ignorance of workout options, and the belief that the borrower can fix the problem by themselves. For example, a recent survey of delinquent Freddie Mac borrowers conducted by Freddie Mac and Roper found that of the 25% of delinquent borrowers who did not contact their mortgage lender:¹³

- 18% believed that they could handle the situation themselves;
- 16% believed that they lacked the money to repay;
- 15% felt fear, embarrassment, or nervousness;
- 12% claimed that they never had difficulty paying their mortgage; and
- 8% believed that there was nothing the mortgage lender could do to help.

Finally, the majority (57%) of delinquent borrowers were not aware of the options that mortgage lenders could offer a borrower who was behind on their payments. The industry is trying to address these issues by educating borrowers about their options through direct mailings, the internet, non-profit agencies, and national toll-free numbers.

Approaches to Loss Mitigation

As the previous section suggests, establishing a communication channel with a co-operative delinquent borrower is no trivial matter but, under the assumption that this has happened, the servicer can follow two broad default management strategies: **Stay at Home** (home retention workouts) versus **Leave the Home** (voluntary title transfers).¹⁴ Home retention workouts include:

- **Partial reinstatement** – The borrower resumes monthly payments and enters a repayment plan that is up to 12 months in duration for past due amounts. For example, if a borrower has a monthly payment of \$2,000 and is one payment delinquent, the servicer can structure a repayment plan in which the borrower pays \$2,200 per month for 10 months and becomes current at the end of this period.
- **Forbearance** – Contractual suspension or reduction in monthly payments for up to 12 months followed by a repayment plan for the past due amounts.
- **Loan modification** – A permanent change in the terms of the mortgage loan in order to create a mortgage payment schedule that the borrower is better able to afford. Typical loan modifications include rate reductions, capitalization of missed payments, and extension of loan term or some combination thereof. A reduction in the borrower's mortgage rate is probably the most commonly used modification strategy. Some servicers will use a trial modification before making permanent changes to the contract. These trial modifications sometimes require an additional one-time upfront payment. More aggressive loan modifications can include writing down the principal balance of the loan by 10%-20% in order to lower the borrower's payment burden.
- **Partial claim** – This is available to FHA borrowers who are at least 4 months but less

¹³ The Roper survey is available on Freddie Mac's web-site.

¹⁴ Of course, the lender could also just decide to straightaway pursue foreclosure but that would not be considered loss mitigation and would instead be classified as the borrower leaving the home involuntarily.

than 12 months delinquent on their mortgage. Under the assumption that the borrower is able to begin making regular payments, HUD will give the borrower an interest-free loan for the delinquent amount. This loan is payable whenever the homeowner prepays their original mortgage for any reason. Fannie Mae has recently introduced a program called **HomeSaver Advance** which shares many of the features of the FHA program.¹⁵

- **Short refinancing** – The borrower is allowed to refinance into a mortgage that is smaller than the amount currently outstanding. Essentially, the lender is forgiving the difference between the old mortgage and the new one. In some cases, the lender may ask the borrower to take out an unsecured loan that is equal to the forgiven amount.

In the case of partial reinstatement/forbearance plans, the loan is typically reported as delinquent to the trust. However, in the case of permanent loan modifications, the loan is usually reported as current if the borrower meets the terms of the modification. Identifying borrowers who have the economic ability to pay and would benefit from such workout strategies is critical. Similarly, it is important that such loan modifications are utilized to minimize losses, not mask underperformance. Also, note that there are some constraints on when loans can be modified if these loans have been securitized as mortgage-backed securities and sold to various investors. These securities are typically issued using a tax-preferred vehicle like the Real Estate Mortgage Investment Conduit (REMIC). In order to maintain this tax-preferred status, which is absolutely critical for the investors of these securities, there are some strict restrictions that must be met on an ongoing basis. These restrictions and any additional restrictions that try to protect the interests of the investors are typically codified in a legal contract (the REMIC Pooling and Servicing Agreement (PSA)) between the trustee and the master servicer.¹⁶

If home retention loss mitigation attempts fail or foreclosure is deemed inevitable due to the distressed financial situation of the borrower, voluntary title transfer workouts that involve loss of the home but avoid costly foreclosure proceedings and foreclosure sales are attempted. Such workouts reduce the high carrying costs, accrued interest, and other expenses that comprise the largest components of foreclosure related liquidations. Voluntary title transfer workouts include:

- **Deed-in-lieu** – The borrower gives up ownership of the property in exchange for cancellation of the mortgage. “Cash-for-keys” settlements are similar but involve a payment to the borrower to vacate the premises.
- **Short sale (short payoff, pre-foreclosure sale)** – This involves a negotiated and lender-approved sale of the property where the borrower has found a buyer for the property but the sale proceeds are less than the outstanding balance of the loan. The borrower may be asked to make up some of the loss based on their financial standing.
- **Mortgage assumption** – Assumption of the property and mortgage by a qualified applicant.

Even if loss mitigation techniques are fully utilized, some properties will inevitably be

¹⁵ See our Trading Desk Strategy Weekly from March 7th, 2008 for details.

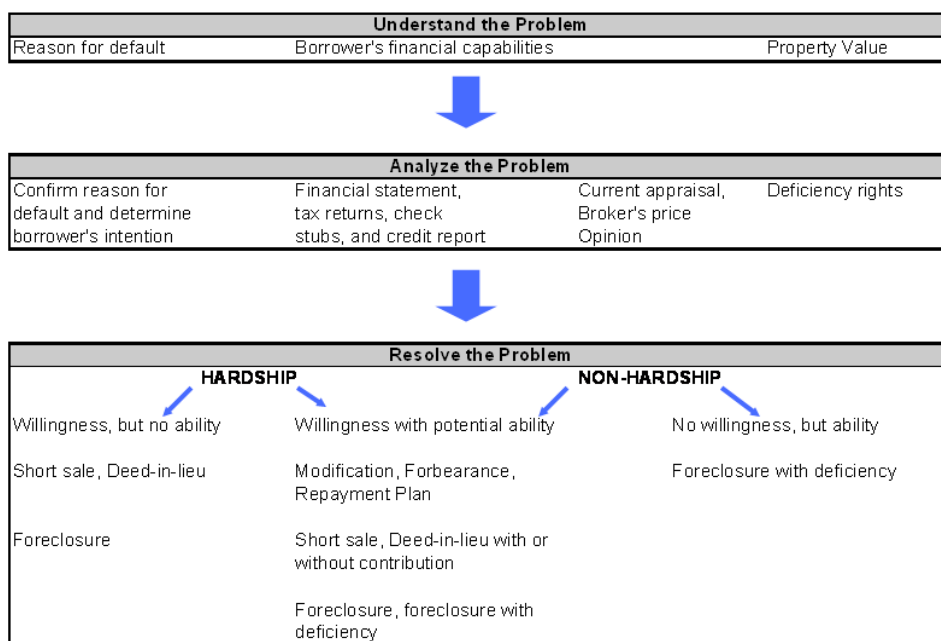
¹⁶ Our subprime primer reviews the legal restrictions with respect to loan modifications based on the PSAs corresponding to the 60 reference deals belonging to the ABX indices.

liquidated through foreclosure sales. Therefore, timeline management in the foreclosure process is also an important component to minimizing losses. In general, the longer the timeline to liquidation, the higher the ultimate costs of foreclosure. Therefore, effective servicers are quick to initiate the foreclosure process while simultaneously pursuing loss mitigation techniques.

Picking an Approach to Loss Mitigation

Given this lengthy list of alternatives, how does a servicer decide which technique to use? Specifically, when should a servicer use a “Stay at Home” technique versus a “Leave Home” technique? There is no cut-and-dried answer to this question as it involves looking at a number of different tradeoffs and in general the answer is very borrower dependent. Figure 3.1 shows a typical “decision tree” a servicer might look at while choosing the appropriate workout option. The decision tree summarizes the choices faced by the servicer and underscores the importance of divining the hardship status of the borrower along with their **ability** and **willingness** to repay the loan. The choice of different loss mitigation options once the servicer decided to go down a particular branch of the tree corresponds to a detailed analysis of the borrower’s cash flow position including what monthly payment they can afford, their income prospects etc.

Figure 3.1: Sample Decision Tree for Workout Resolutions



Source: Mortgage Guaranty Insurance Corporation

Why Loss Mitigation is Not a Panacea

As we have emphasized throughout our discussion, on average, a workout saves the lender money versus the alternative of foreclosure. Still, the very important point to keep in mind is that every single borrower who is put into a loss mitigation plan will not cure and, in fact,

many will **re-default**. For those that re-default, the loss severity rate can be much higher than if the servicer had foreclosed the property at the very onset of serious delinquency, especially in an environment of declining home prices. Note also that in a failed workout the borrower is even further in arrears than before. All of this means that the servicer needs to keep careful track of the costs associated with the success and failure of various loss mitigation techniques and evaluate the borrower's probability of success with respect to a particular technique before proceeding. To develop this idea a little more, let **E(Workout)** and **E(Immediate Foreclosure)** be the expected costs associated with a particular workout option and an immediate foreclosure respectively. Clearly, we want **E(Workout) < E(Immediate Foreclosure)** in order to proceed with the workout. If p denotes the average borrower's probability of success (in other words the probability that they "cure"), then:

$$E(\text{Workout}) = p * \text{Cost of a successful workout} + (1 - p) * \text{Cost of unsuccessful workout}$$

The value of p that makes **E(Workout) = E(Immediate Foreclosure)** is known as the **breakeven workout probability** and in general we want to pursue loss mitigation strategies with borrowers whose probability of success is greater than this number. The breakeven workout probability in conjunction with a borrower's individual probability of success is used in the industry by servicers, insurers and other participants to assess the cost-benefit tradeoffs of the workout process.

IV. ANATOMY OF MORTGAGE DEFAULTS

Introduction

If loss mitigation fails or the servicer is unable to reach the borrower, foreclosure becomes an inevitability. In this section, we flesh out some of the details associated with foreclosure process and also clarify the difference between foreclosure and the various notions of mortgage default even though we have somewhat imprecisely been using all these terms interchangeably so far.

As we mentioned in our discussion of loss mitigation, although borrowers are technically in default of their mortgage contract after missing even one payment (**contractual default**), lenders do not attempt to start the foreclosure process in this situation. In practice, commencing foreclosure proceedings requires a couple of months of delinquency on part of the borrower followed by a failure to contact them, or an inability to come to terms on a workout plan to repay the loan, or unsatisfactory performance by the borrower on a current workout plan.¹⁷ One moment of reckoning in the foreclosure process arrives when the borrower's property is auctioned off (the **foreclosure sale**) to repay the loan although, as we will discuss later, the borrower still has an opportunity to redeem the property after the sale for a limited period of time in some states. The foreclosure sale will not result in repayment of the outstanding loan if the winner of the auction is the lender/servicer, which is what often happens in practice. If the lender/servicer is the highest bidder at the auction, the property becomes part of their "real estate owned" (or **REO**) portfolio and the liquidation of the REO property is what will lead to repayment of the loan.

The above paragraph is perhaps a long-winded way of saying that in addition to the contractual definition of default (a borrower misses a certain number of monthly payments), we can also say that a default event occurs when the property of a delinquent borrower is liquidated *and* the proceeds are used to satisfy as much of the outstanding debt as possible. This is a more accurate measurement than just looking at foreclosure starts because all foreclosure proceedings will not necessarily be carried through.¹⁸ Furthermore, while the majority of all mortgage defaults as defined above will occur through the foreclosure sale process, a significant number of them can also occur through short sales (or, less commonly, through assumptions and deeds-in-lieu of foreclosure), as discussed in our section on loss mitigation.

Unfortunately, there are some practical issues associated with implementing our proposed measurement of default because of the level of disclosure in publicly available loan-level data sets (such as Loan Performance) that report on the credit performance of the mortgage loans underlying MBS deals. The issue is that these data sets only publish the delinquency status of a loan and whether the loan has paid off or not.¹⁹ Thus, in practice, we define a **default** as occurring when a loan that was more than 3 monthly payments behind (including being in the foreclosure or REO stage) pays off. Our definition of default does not require the presence of a loss, so it is possible for a loan to default and to still recover all the principal (this might happen in a short sale, for example).

To summarize, although the two terms are often used interchangeably, mortgage default and foreclosure are not one and the same thing. Foreclosure is a legal process which extinguishes the borrower's rights to the property through transferring the title from the borrower to the new

¹⁷ The lender can also initiate foreclosure proceedings if the borrower is not paying property taxes, not maintaining their insurance, or is allowing environmental damage to their property.

¹⁸ A similar objection applies to using contractual defaults to measure default events.

¹⁹ In other words, the data sets do not reveal *why* (turnover, refinancing, default) a loan has paid off.

owner of the property, with the transfer usually being accomplished through a property sale.²⁰ Mortgage default is a more general idea and corresponds to any situation (short sale, foreclosure sale etc.) where a delinquent borrower's property is liquidated at a loss.

Having said all of this, our initial focus in this section is on studying the typical foreclosure process because some of the loss mitigation-related activities that lead to defaults have already been covered in our treatment of loss mitigation and also because most defaults are realized through the foreclosure process. On the other hand, while all of this material tells us how default happens, it still leaves us in the dark as to why default happens and thus the section concludes with a brief review of the economic determinants of defaults.

The Foreclosure Process

Before we get into the details, it is useful to get a couple of 10,000 foot views of the landscape. What happens in the foreclosure process is codified through law and there are three general facts to keep in mind about this body of law:

- Foreclosure law is determined by the states;²¹
- There is a considerable amount of variation in state foreclosure law; and,
- State foreclosure laws are superseded by the Federal Bankruptcy code.

We will go into a fair amount of detail on foreclosure law and how it interacts with the different parts of the foreclosure process but, before we do so, it is once again useful to get a high-level overview. We can divide the typical foreclosure process into three segments:

- **Prior to the foreclosure sale ("Pre-foreclosure").** This is the period between the notifying the borrower that foreclosure proceedings have been initiated and the actual foreclosure sale.
- **The foreclosure sale.** This short period (1 day) corresponds to the sale of the house at a public auction to settle all existing claims.
- **After the foreclosure sale.** In most cases, the lender will be the winning bid at the foreclosure auction and end up owning the house. The house will be part of their REO inventory until it can be sold. In other cases, the house will be sold to a third-party investor.

Now, for the details.

Pre- Foreclosure

To repeat ourselves, the foreclosure process starts when the lender has determined that the borrower's delinquency status is serious enough that there is a real possibility the borrower will not be able to honor the mortgage contract. At this juncture, the path a lender follows depends upon state-level law. There are principally two types of foreclosure actions that can be followed in the states: **power-of-sale** (also known as **nonjudicial** foreclosure) and **judicial foreclosure**. Judicial foreclosure is available in all states and is the only foreclosure option available in nearly half of these states. Stated differently, a slight majority of states allow both types of foreclosure actions but nearly half of all states only allow for judicial foreclosure. The

²⁰ We say "usually" because for states which follow a **strict foreclosure** process (Connecticut, Vermont), the lien transfer takes place through the court granting the lender title to the property. No sale of the property takes place.

²¹ Unless the government has countervailing interests. The contexts in which the federal pre-emption of state level property laws applies is still not well understood. See Capone (1996) for details on this issue.

key features of both these two types of foreclosure actions in the pre-foreclosure stage can be described as follows:

- **Notice of Intent (NOI).** This is an official notification that foreclosure proceedings have begun and is sent to the borrower by the servicer. NOIs are also sent to junior lien-holders and filed in the county court. In judicial foreclosures, the formal notice that starts the foreclosure process is called a **lis pendens** (“suit pending”).
- **Hearing.** In a judicial foreclosure, a judge hears the court case that the servicer has filed against the borrower. If the judge rules for the lender, the date for a court-supervised sale of the house is set. In a power-of-sale foreclosure, the servicer or trustee usually proceeds directly with the sale.²²
- **Notice of foreclosure sale.** The foreclosure sale must be advertized for a minimum amount of time in both the judicial action and power-of-sale process through a posting or a newspaper ad.

A very important right the borrower possesses in the pre-foreclosure stage is called the **equity of redemption**. This right means that at any point prior to the foreclosure sale, the borrower can stop the foreclosure process by paying the amount of outstanding mortgage debt (plus late payments and legal fees) in full. This right is available to the borrower in all states. However, the practical utility of it is somewhat circumscribed by the fact that the borrower has to come up with a significant amount of funds and/or a new financing arrangement and may not have much time to accomplish all of this in the relatively speedy power-of-sale process. There is also another form of legal protection available to the borrower before the foreclosure sale – **arrearages legislation**. This basically says that the borrower can reinstate their mortgage (cure the default) by making up the missed payments (plus other penalty fees). Some states limit the number of times a borrower can reinstate a loan.

The Foreclosure Sale

A foreclosure sale has some unusual features and one aspect that jumps out to a first-time observer is how disadvantaged any non-servicer-related bidder (“third-party bidder”) is in the sale process. For example, the third-party bidder cannot conduct any due-diligence by inspecting the property before the bid and in addition has to face the constraints that all sales are final, sales have non-negotiable terms, and sales are without recourse. Equally importantly, potential buyers are required to have lined up financing and have to put down significant cash deposits of 10%-20% of the property price on the auction date (and pay the remainder within a short period of time).²³

The sale price is determined by an “English auction” bidding process, which the servicer is always a part of. This type of auction can be described as follows:²⁴ (1) Buyers congregate in a single location; (2) Buyers call out bids so that each buyer is aware of other bids; (3) Each successive bid has to be higher than the one before; (4) An individual may bid several times; and (5) The sale is awarded to the highest bidder.

²² The trustee or servicer’s right to sell the property without going through a court gives them “power-of-sale” and thus explains the naming convention for this foreclosure method.

²³ For further details, see Grant S. Nelson and Dale A. Whitman, *Reforming Foreclosure: The Uniform Nonjudicial Foreclosure Act*, Duke Law Journal, Vol. 53, Number 5, March 2004.

²⁴ See Nelson and Whitman (2004).

Given the constraints faced by the third-party bidder, it is not surprising that more often than not the servicer is the winning bid in foreclosure sales.²⁵ The amount bid by the servicer is typically prescribed by the entity that is long the credit risk on the mortgage loan. For example, on loans guaranteed by Freddie Mac, the company requires that servicers start bidding at the statutory minimum level for a state (usually 2/3rds of the property's value) until the bid reaches the "90-day fair market value" or the amount of outstanding debt, whichever is less. The 90-day fair market value is the amount expected to be realized from the sale of the property in the next 2 to 3 months, assuming no further discount is necessary.²⁶

The foreclosure sale proceeds are used to pay off all the mortgage holders in order of their priority. In the unlikely event that there is a surplus (most borrowers would not default if they had equity in their property), it goes to the borrower. In many cases the sale amount will fall short of the outstanding loan amount (plus foreclosure-related expenses). The difference between the outstanding mortgage debt and the foreclosure sale price plus default-related costs is called the **deficiency amount**. In theory, the borrower is personally liable for this amount in many states although, as Appendix D explains, lenders will rarely pursue borrowers. The borrower also potentially owes taxes on the amount of forgiven debt (see Appendix G).

After the Foreclosure Sale

Even though the property has been sold, the borrower still has some residual rights. First, they could potentially stay rent-free as "hostile occupants" until they are evicted. Second, borrowers have a **statutory right of redemption** in judicial foreclosures (this right is not typically available in power-of-sale foreclosures). This right gives the borrower the ability to buy back the house by paying the same price as the winning bid (the borrower does not have to bid the amount of the foreclosed debt) within some fixed period of time from the foreclosure sale date. This time period can vary from 10 days to 12 months depending upon the state (see Appendix D).

If a third-party did not buy the property at the foreclosure sale, the property goes into the lender/servicer's inventory and gets liquidated through the **REO Sale** process. REO sales are a little more friendly to third-party investors than the foreclosure sale process but there are still some significant restrictions to deal with. Investors have to be pre-approved for a loan before they are allowed to bid, are not allowed to conduct an inspection (this is a significant issue because servicers will have often turned off electricity, plumbing, and mechanical systems in the house in order to lower their carrying costs), and all sales are without recourse. REO properties are usually marketed through real estate brokers.

Bankruptcy and Foreclosure

The inter-relationship between bankruptcy and foreclosure comes from the fact that the borrower can freeze the foreclosure process through the **automatic stay** by filing for bankruptcy. The details behind how this happens and the strategic considerations involved for a borrower in choosing to file for bankruptcy are explored in considerable detail in Appendix F. In general, filing for bankruptcy increases the length of time spent in the foreclosure process.

²⁵ Empirical research on this topic, while somewhat dated, suggests that the lender is the successful bidder 80%-90% of the time. See Nelson and Whitman (2004).

²⁶ For further details on the bidding requirements for FHA, VA, RHS, Fannie Mae and Freddie Mac, see the GAO report, *Opportunities to Improve Federal Foreclosure and Property Sale Processes*, GAO-02-305 (April 2002).

Foreclosure Timelines and the Cost of the Foreclosure Process

Fundamentally, investors care about the foreclosure process and all the minutiae associated with it because it relates to their recovery of principal. Section VI on loss severity will explore this issue in a lot more depth so we will content ourselves with making the following general observations here:

- The longer the foreclosure process, the greater the expenses incurred and thus the greater the loss realized on the default
- There is a greater loss severity associated with REO sales as compared to short sales and foreclosure sales

To amplify a little on the first point, a mortgage investor will usually prefer a power-of-sale foreclosure process over a judicial foreclosure since the latter typically lasts 3 months to 5 years, while a power-of-sale takes somewhere between 1 to 8 months to complete. Consequently, given that some states do not offer a power-of-sale, the time spent in the foreclosure process can vary significantly by state. These state-wide differences in timelines (and other expenses) can be captured through a single number – the cost of foreclosing in a state. Figure 4.1 provides a snapshot of how this cost compares for the top mortgage states.

Figure 4.1: Foreclosure Timelines for Selected Mortgage States

State	Foreclosure Type	Avg. #Days from DDLPI to FCL Sale ¹	Avg. Cost from DDLPI to FCL Sale Relative to US Avg. ²
Arizona	POS	253	68%
California	POS	268	171%
Colorado	POS	339	101%
Florida	Judicial	326	96%
Georgia	POS	241	69%
Illinois	Judicial	398	112%
Maryland	POS	274	65%
Massachusetts	POS	263	142%
Michigan	POS	380	70%
Minnesota	POS	425	82%
Nevada	POS	283	118%
New Jersey	Judicial	436	224%
New York	Judicial	392	118%
North Carolina	POS	281	72%
Ohio	Judicial	480	107%
Oregon	POS	369	104%
Pennsylvania	Judicial	453	110%
Texas	POS	254	93%
Virginia	POS	213	63%
Washington	POS	299	98%

¹ Based on a sample of Freddie Mac REO property acquisitions in 2007.

² Based on a sample of Freddie Mac REO property acquisitions prior to Sep. 30, 2007. A value of 100% means that the average pre-foreclosure cost in that state is equal to the national average cost. Costs include interest expenses prior to foreclosure sale.

DDLPI = Due date of last paid installment

Source: Freddie Mac

Determinants of Mortgage Default

Thus far we have addressed the “how” of mortgage foreclosure and default but not the “why.” Mortgage delinquency and mortgage defaults will clearly share the same underlying causes (see Figure 2.1) but, as a first step towards successfully building models, we will need to develop a more conceptual view of the underlying economic determinants of defaults.

The first step towards this conceptual understanding comes from viewing the homeowner’s right to default as equivalent to them owning a put option on the house. In down to earth terms, a necessary condition for the borrower to default is if the value of the house is less than the market value of the mortgage.²⁷ The condition is not a sufficient condition for default because the borrower has to weigh the high costs associated with foreclosure that we had previously listed: moving expenses, the cost of a damaged credit record, the expected cost of deficiency judgments etc. Consequently, in addition to negative equity, borrowers

²⁷ In this situation, the homeowner is said to have “negative equity.” The market value of the mortgage is an important component of the homeowner’s decision to default because a discount mortgage is more valuable to the homeowner than a premium.

typically have to be subject to some **trigger event** in order to default – a shock to their income resulting from sustained unemployment, a divorce, an unexpected increase in health-related costs or some other form of liability.

Patterns in Default-related Prepayments

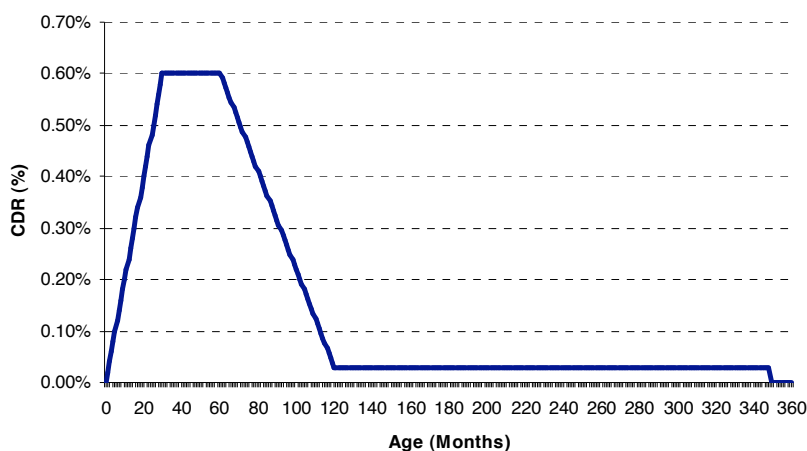
The initial attempts at modeling default rates did not consider the economic determinants listed above and simply tried to capture the distinctive dependence of default rates on the age of the mortgage. The general pattern can be described as follows. Default rates start out low since lenders will not underwrite borrowers who they suspect will default immediately. Also, since mortgages with LTVs of 100% or greater are still fairly rare in the U.S., most borrowers typically have positive equity at origination. We start to see default rates ramp up gradually in the first two or three years after origination since it is over this period that borrowers are most vulnerable to losing all their equity if home values decline. After this, default rates start tapering off five or more years after origination since increases in home prices and an increasing amortization effect decreases the likelihood of negative equity.

The SDA Model for Defaults

The **SDA (Standard Default Assumption)** model is a benchmark curve for default analysis developed by the Bond Market Association that tries to capture the pattern described above. The curve is presented in Figure 4.2 and is given by the following recipe:

- Beginning with a CDR of 0.02%, default rates increase by 0.02% CDR per month until month 30, when the peak value of 0.60% is reached;
- Defaults remain at the value for the next 30 months;
- Starting in month 61, defaults decrease by 0.0095% per month until month 120, when the tail value of 0.03% is reached. Defaults remain constant at this level for the remaining life of the security.
- By convention, the last 12 default rates are set to zero, assuming it takes 12 months to liquidate a property.

Figure 4.2: The SDA Curve



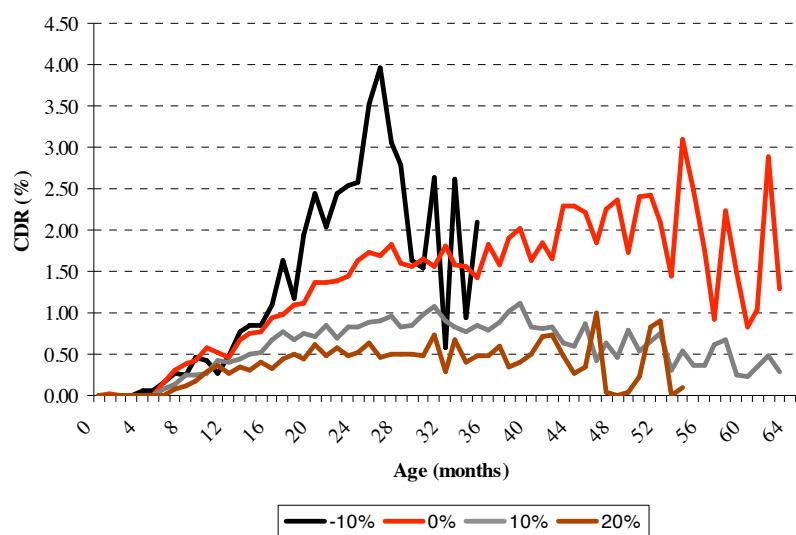
Source: Banc of America Securities

Strengths and Weaknesses of the SDA Model

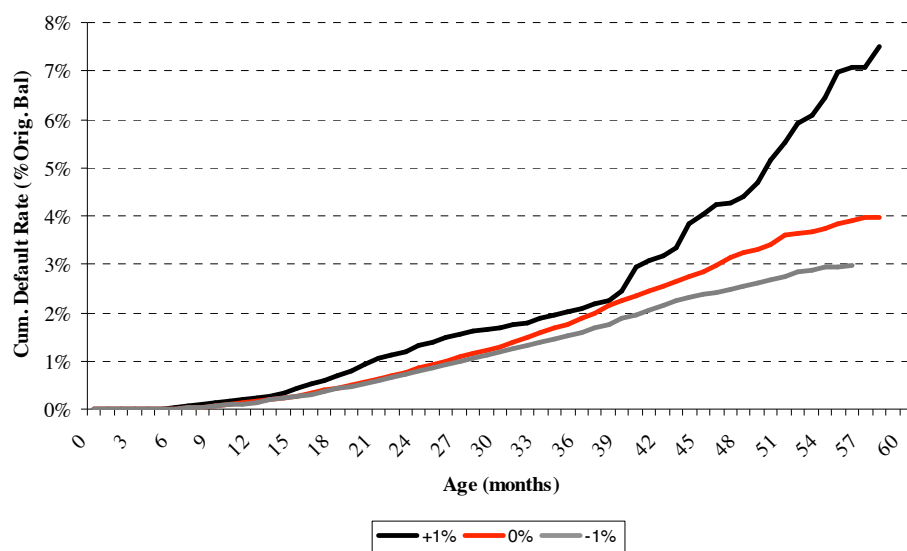
The shape of the SDA curve more or less captures the age-dependent nature of default rates described above. The reason the curve works reasonably well in describing the dynamics of defaults is because, since home prices have generally tended to increase over time, age is strongly correlated with equity growth, which is one of the primary determinants of the decision to default. One of the several weaknesses of the model is of course the fact that the assumed relationship between defaults and aging will tend to break down in a decreasing home price environment.

A more realistic model will reflect the more conceptual perspective on defaults we started with (default as a put option, the importance of negative equity and trigger events). For example, Figure 4.3 shows how mortgage defaults vary as a function of the home price appreciation experienced by borrowers, while Figure 4.4 shows the dependence of defaults on unemployment rates. A default model that incorporates these macroeconomic and microeconomic (the borrower's current LTV, credit score and debt-to-income ratios) variables is presented in the next section.

Figure 4.3: The Dependence of Mortgage Defaults on Home Price Appreciation



Home price appreciation is computed on an annualized basis using Freddie Mac's MSA-level home price indices. The mortgage loans correspond to 30-year fixed-rate alt-A borrowers.
Source: Banc of America Securities, Freddie Mac

Figure 4.4: The Dependence of Mortgage Defaults on Unemployment Rates

Unemployment rates are computed at the MSA level and measured as the total change in unemployment rates between the origination of the loan and the current period.

The mortgage loans correspond to 30-year fixed-rate alt-A borrowers.

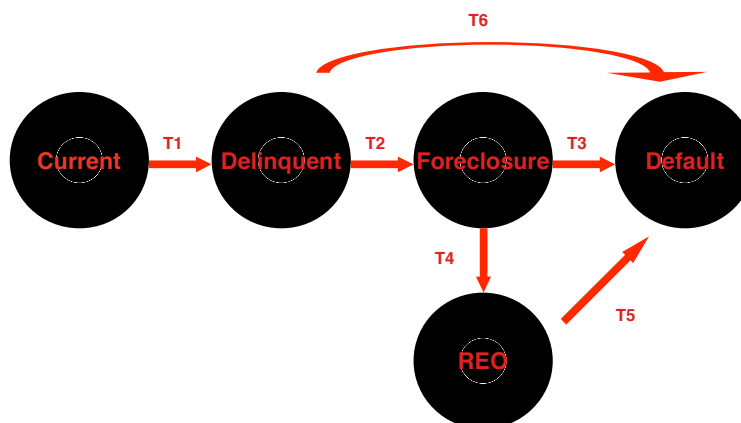
Source: Banc of America Securities, BLS.

V. MODELING THE DELINQUENCY AND DEFAULT PROCESS

An Approach to Modeling Defaults

The skeleton of a useful framework for capturing the dynamics of mortgage delinquency and default is presented in Figure 5.1. The framework is inspired by Figure 1.1 and our subsequent discussion in that it conceptualizes the life of a mortgage as consisting of various **payment states** (**Current**, **30-day Delinquent**, **60-day Delinquent** etc.) with different sequences of paths from one state to another. To simplify matters for the purposes of exposition, the figure assumes that a mortgage can only be in one of the five states below and the arrows connecting the different states (T1 through T6) represent the only possible transitions between states. Notice that the payment state transitions model the various processes we have talked about in the primer. For example, a loan going from **Current** to **Delinquent** to **Default** corresponds to a short sale.²⁸ Similarly, the path **Current** -> **Delinquent** -> **Foreclosure** -> **REO** -> **Default** corresponds to the typical foreclosure process. Now, if we can determine the probability of a loan transitioning from one state to another, we can then use these transition probabilities to forecast, on a month-by-month basis, the rate (frequency) with which a pool of mortgages will “roll” from one state to another. Once we know this, it is easy to calculate the voluntary prepayment rate, default rate, delinquency rate and other metrics for the pool. This framework is known as a **roll-rate** model. Our goal in this section is to go into some of the details of how roll-rate models are constructed and used in practice. We also review some of the practical difficulties associated with estimating these models and interpreting their results.

Figure 5.1: Examples of Loan Payment Status Transitions



Source: Banc of America Securities LLC.

Fundamentals of Roll-Rate Modeling

To flesh out some of the details associated with roll-rate models, let's start by classifying the different (distinct and mutually exclusive) states that a mortgage can be in. Here is one set of possibilities:

²⁸ Recall our definition of mortgage default presented in Section IV.

- Current
- Current and Bankrupt
- 30 days Delinquent
- 30 days Delinquent and Bankrupt
- 60 days Delinquent (including Bankruptcy)
- 90+ days Delinquent (including Bankruptcy)
- Foreclosure (including Bankruptcy)
- REO
- Voluntary payoff (Prepayment)
- Involuntary payoff (Default)

To once again illustrate some of the ideas we presented above, let's express the mortgage default process in terms of the above states :

- The borrower misses a monthly payment and the loan becomes delinquent: The mortgage status changes from **Current** to **30 Days Delinquent**.
- The servicer sends a notice to the homeowner for the missing payment (s): The mortgage status starts to reflect increasing levels of delinquency: **60 Days Delinquent**, **90+ Days Delinquent**, etc.
- After 90 days, the servicer begins the foreclosure process. At the same time, the servicer will employ various loss mitigation techniques such as partial reinstatements, forbearance, loan modifications, deed-in-lieu of foreclosure, short sale, etc: The mortgage status changes from **90+ Days Delinquent** to **Foreclosure**.
- If the above-mentioned loss mitigation techniques fail, the property is then auctioned off through a foreclosure sale: The mortgage status stays in **Foreclosure**.
- If there are no buyers after the foreclosure auction, the property enters REO and is liquidated by the servicer. The mortgages status changes from **Foreclosure** to **REO** when the property enters REO. A liquidation would result in the property going into **Default** status.

What we want our model to do is to calculate the probability of a loan transitioning from any of these states to another. To see how we would represent these probabilities, take a look at Figure 5.2 which shows the transition matrix for a “toy” roll-rate model with just four states: **Current (C)**, **Delinquent (D)**, **Voluntary Payoff (V)**, and **Default (L)**. The first row of this matrix can be interpreted as saying that the probability of a current loan remaining current is 95%, the probability that it transitions to delinquency is 1%, and the probability that it prepays is 4%. Note that the matrix in Figure 5.2 also makes the reasonable assumption that a mortgage cannot move directly from a **Current** status to a **Default** status, hence the 0% probability associated with this transition. We can only go from a **C** or a **D** state to some other state because **V** and **L** are **absorbing** states – once a mortgage has prepaid or defaulted, it cannot go to any other state.

Figure 5.2: Sample Transition Probabilities for a Simple Roll-Rate Model

From	To			
	C	D	V	L
C	95%	1%	4%	0%
D	20%	75%	3%	2%

Source: Banc of America Securities LLC

To understand how we use this roll-rate matrix for forecasting purposes, let's walk through another example. Assume that we have a collection of loans with 80% of them being current and 20% being delinquent. This is the starting state at time $t = 0$. Based on the transition probabilities in Figure 5.2, one month from now at $t = 1$:

- $3.8\% = (0.80 * 4\%) + (0.20 * 3\%)$ of the $t = 0$ balance will be paid off voluntarily (37.2% CPR)
- $0.4\% = (0.80 * 0\%) + (0.20 * 2\%)$ of the $t = 0$ balance will default (4.7% CDR)
- $80\% = (0.80 * 95\%) + (0.20 * 20\%)$ of the $t = 0$ balance will be current
- $15.8\% = (0.80 * 1\%) + (0.20 * 75\%)$ of the $t = 0$ balance will be delinquent
- $16.5\% = 15.8\% / (80\% + 15.8\%)$ of the $t = 1$ balance will be delinquent
- $83.5\% = 80\% / (80\% + 15.8\%)$ of the $t = 1$ balance will be current

In practice, we use the model at the loan-level and apply it to the dollar balance of the loan. For example, assume that we have a loan that is current and that it has a balance of \$100,000. Since we know the current state of the loan, we have a special case of the above example with 100% of the loan balance in the C state. Now, using the same calculation logic as above, in the next month \$95,000 will remain current, \$1,000 will become delinquent, and \$4,000 will payoff. In other words, we are treating each \$1 of the balance as a separate loan.

If we opt for a slightly richer collection of payment status than those presented in Figures 5.1 and 5.2, it should be clear from the above example that, for any collection of loans, we can use a roll-rate model to forecast the following mortgage performance measures on a month-by-month basis:

- Voluntary prepayments;
- 60+ Delinquency rates; and,
- Defaults (involuntary prepayments).

Finally, given a projected vector of defaults and a model for **loss severity**, we can also compute a vector of expected losses (month by month) on mortgage loans.

Readers who are used to the pool-level models used to forecast prepayments on Agency MBS will note that loan-level roll-rate models are somewhat more complicated than standard prepayment models, with the latter essentially being a special case of a mortgage roll-rate model with only two states: **Current** and **PaidOff**.

²⁹ See our primer: *Credit Structuring Techniques in Prime Non-Agency MBS*, February 23rd, 2007.

At this point, one might well ask: Why do we need to be able to model the transition probabilities between so many different mortgage states? While in practice the number of payment statuses can be simplified for the purposes of computational tractability, there are some important reasons to try to capture the various degrees of delinquency by having more states. Specifically:

1. The distribution of payment states of a pool of loans – Current, one payment delinquent, two payments delinquent etc. (the **delinquency pipeline**) – is an important predictor of future defaults and losses for the pool. In other words, for mortgage loans, past performance *is* predictive of the future.
2. The **triggers** on private-label MBSs are checks on the delinquency rate and cumulative losses in a pool of mortgages that ensure that senior tranches remain protected in the event of a credit meltdown. Triggers protect senior tranches by preventing or suspending stepdowns when they are tripped. For example, a typical trigger test is to suspend prepayments to subordinate tranches and funnel these to the seniors when the 6-month average of the 60+ Day Delinquent Balance is greater than 50% of the balance of all outstanding subordinates.²⁹
3. The time spent in delinquency (**delinquency history**) is a powerful predictor of future performance.

So far, so good. But, even at a superficial level, it should be evident that the roll-rate model described above is much too simplistic. The assumption that roll-rate probabilities remain constant over different borrower attributes and economic environments is clearly unrealistic. For example, the transition from **Current** to **Delinquent** will clearly depend upon the borrower's economic circumstances including whether they have recently experienced some interruption in their income. We now go through some of the specifics of our framework for modeling mortgage credit, which is much more realistic than the simple example presented in Figure 5.2.

Mortgage Transitions in Practice

For practical considerations and reasons of computational efficiency, our implementation of the roll-rate model reduces the number of transitions modeled by assuming that the mortgage can only be in one of the following payment statuses:

- C: Current or 30 days Delinquent
- B: Current and Bankrupt
- D: Delinquent (for 60 days or more)
- F: Foreclosure (including Bankrupt borrowers with 30 or more days of delinquency)
- R: REO
- V: Voluntary payoff
- L: Involuntary payoff

In Figure 5.3, we show the matrix of transition probabilities that we model. An empty cell in the matrix indicates a transition probability of zero. Cells marked with “I” are derived from other entries in the same row since the sum total for each row is 100%. So, for instance, our model allows transitions from **C** to **C**, **C** to **D**, **C** to **B**, and **C** to **V**. But, there is no direct transition possible from the **Current** to **Default** stage.

Figure 5.3: Modeled Transitions

	To						
From	C	D	F	R	B	V	L
C	I	X			X	X	
D	X	I	X	X		X	X
F	X	X	I	X	X		X
R				I			X
B	X		X		I	X	

Source: Banc of America Securities LLC

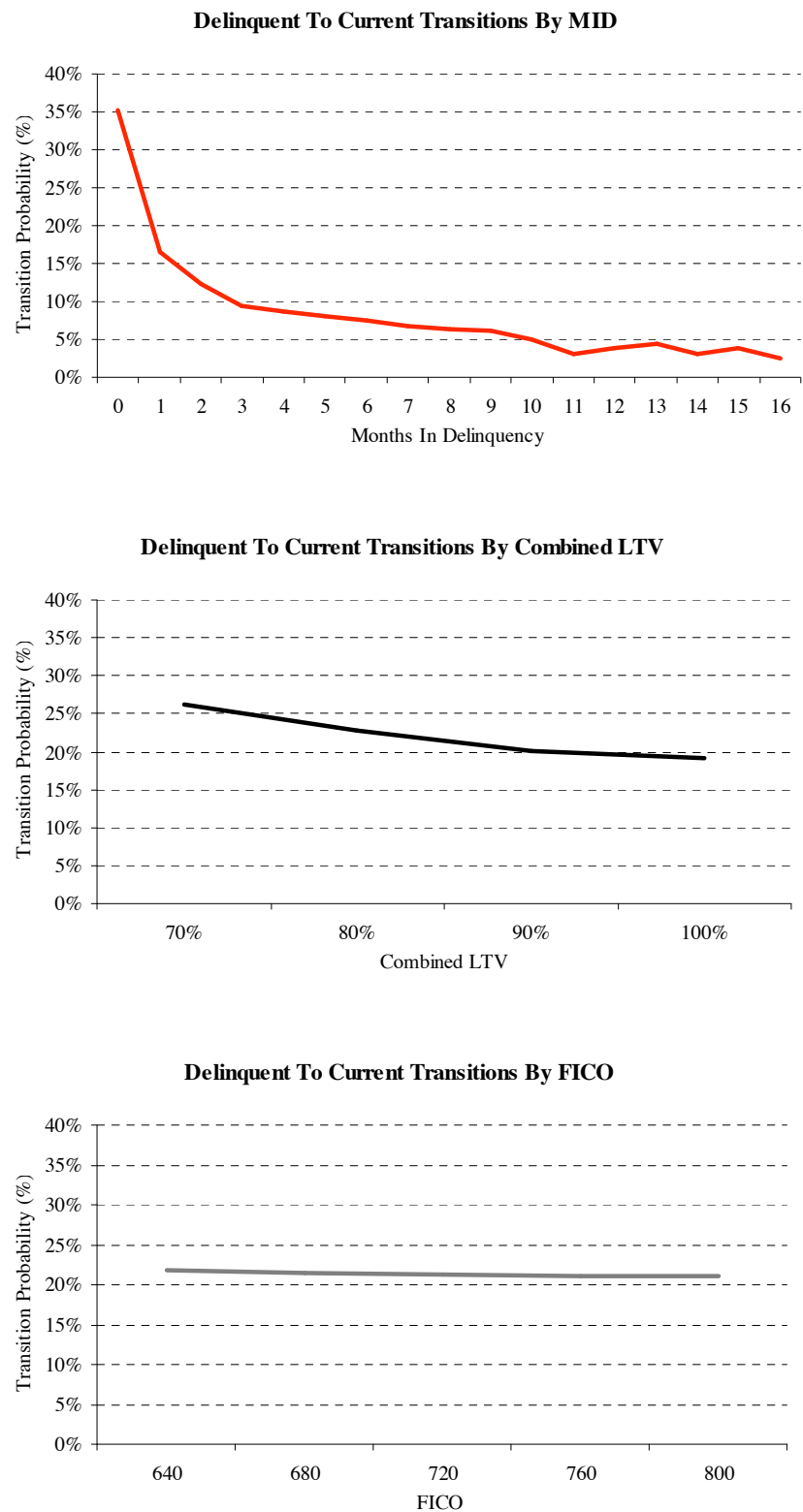
In practice, each transition probability depends upon an array of factors that are traditionally used in prepayment models including:

- Incentive
- Loan Age
- Loan Size
- FICO
- Combined LTV
- Occupancy Status: Owner-occupied or Investor
- Loan Purpose: Purchase or Refinance
- Documentation: Full, Alternative, None
- Home Price Appreciation (HPA)
- Unemployment
- State-level foreclosure laws (**Judicial** versus **Power-of-Sale**)

Besides the conventional list of variables above, we can also use information that we don't ordinarily have access to in pool-level models such as the past history of payment statuses on the loan. As mentioned above, transition probabilities also depend upon a loan's **delinquency history**. There are several ways to summarize the delinquency history of a loan. One way is to count the **months-in-delinquency (MID)**: For a loan whose status is one of D, F, or B, we count the number of months since the loan was last current (C).

The more variables a model has, the harder it is to calibrate and disentangle the net effects of different factors. Thus, one goal of the estimation process is to whittle down the list of factors influencing a specific transition probability to as small a number as possible. To illustrate, an empirical study of the **Delinquent to Current** transition probability for fixed-rate 30-year Alt-A loans shows the strongest dependence on MID, the initial CLTV of the mortgage, and HPA. The results are consistent with our economic intuition. The longer a loan stays delinquent, the less likely it is that it will become current again. The initial CLTV reflects how leveraged the borrower is, and HPA determines the borrower's equity position. Figure 5.4 puts some numbers behind these statements. The first graph shows the overall transition probability decreasing from 35%, when a loan just becomes delinquent (MID = 0), to about 5% when MID is 16 months or more. The second graph shows that CLTV is also important, but less so than MID. Finally, the loan's initial FICO plays little to no role in influencing the **D** to **C** transition.

Figure 5.4: The Dependence of *Delinquent to Current* Transitions on Loan Attributes



Source: Banc of America Securities LLC

Complications

As always, the devil is in the details and clearly there are enough details to occupy several devils here. There are 17 different transition probabilities to model in Figure 5.3 and each of these depend upon several different factors. These factors may be correlated to each other. The dependence of the transition probabilities on these factors may be a complex one (For example, look at the very non-linear relationship between **Delinquent** to **Current** transitions and **MID** in Figure 5.4). Even a well-estimated roll-rate model is not particularly easy to use because of the computational complexities associated with running each loan in a deal over several scenarios. Finally, perhaps the most challenging aspect of roll-rate models is translating the inevitable structural changes in the mortgage market environment to the model. For example, how does an increase in the transaction costs associated with refinancing affect the **C** to **V** probability? Should it affect the **C** to **D** probability?

In models, simplicity is often a virtue and this is the chief drawback associated with roll-rate models. Having said that, the dependence of mortgage credit performance on payment history and the ability of these models to capture a host of other loan-level information (loan-level FICOs, CLTVs, HPIs) makes them a key tool in the arsenal of a mortgage credit analyst.

VI. CALCULATING AND MODELING LOSS SEVERITY

Introduction

When a borrower does default on their loan, **Loss Severity (LS)**, determines how much of the unpaid loan principal is *not* recovered.³⁰ LS is frequently greater than zero in default situations because even though the lender will try to recover their principal through the foreclosure process, the value of the home is typically less than the unpaid principal balance (UPB) of the loan by the time we get to this situation. This is because, as we argued in Section IV, a borrower with positive equity in their home would not choose to default. Furthermore, there are several costs incurred by the lender in the default process which make it even tougher to break even.

In terms of definitions, loss severity is usually expressed in terms of the unpaid principal balance of the loan (or sometimes the original principal balance) as per Figure 6.1. For forecasting purposes, we usually prefer expressing it as a function of the unpaid principal balance (UPB) of the mortgage loan at the time of default. Applying loss severity to the defaulted balance give us the total loss experienced on the loan.

Figure 6.1: The Loss Severity Calculation

$$LS = \frac{\text{Loss}}{\text{Unpaid Principal Balance}} \times 100$$

Source: Banc of America Securities LLC.

The Components of Loss Severity

Our brief discussion has already isolated two of the components of loss severity: (1) The difference between the outstanding amount of the loan and what the lender obtains from selling the house; and (2) The operational costs associated with the default process. A more thorough categorization of the various components of loss severity is as follows:

- **Base Severity.** The difference between the UPB and the final sale price of the house, expressed as a fraction of the principal balance.
- **Real Estate Commissions.** Amount due to the real estate broker upon sale of the house plus any seller-paid closing costs. The broker amount typically averages 6% of the sale price.
- **Foreclosure Costs.** Legal expenses associated with the foreclosure process, delinquent portions of property and insurance tax payments.
- **Holding Costs.** Property maintenance, property taxes, hazard insurance, utilities.
- **Uncollected (Lost) interest.** Missed interest payments by the borrower.
- **Insurance payments.** Recoveries from mortgage insurance. These payments reduce loss severity.

³⁰ The **Recovery Rate (RR)** is the complement of Loss Severity ($RR = 100\% - LS$) and measures how much of the existing principal balance of the loan is recovered upon default.

Unfortunately, most publicly available data sets of losses on mortgage loans do not have information on the relative contribution of these different components to the total amount of loss severity. The results we do have date from an old study based on a sample of foreclosures in California over the 1992-1994 time period and are summarized in Figure 6.2.³¹ The data underlying the figure consist of 117,000 foreclosure sales that occurred in California between January 1992 and February 1995. All major metropolitan areas are included in the dataset and approximately 95% of the state of California lived in the 30 counties that these sales occurred in. As the figure demonstrates, base loss severity is the most important contributor (~38%) to overall loss severity, followed by uncollected interest (~29%) and foreclosure costs (~19%).³² Note that the figure calculates loss severities as a function of the original principal balance.

Figure 6.2: The Components of Loss Severity

	Fixed	ARM
Base Loss Severity	15.0	15.0
Real Estate Commission	5.3	5.3
Principal Amortization	-1.7	-1.7
Foreclosure Costs	7.6	7.6
Holding Costs	1.3	1.3
Uncollected Interest	11.4	7.6
Estimated Loss Severity	39.0	35.2

Interest based on weighted-average (30-year schedule) rate by date of origination for fixed, and COFI from 1992 through 1994 plus 2.3% for ARMs. Holding costs are 186 days at 2.6% per year.

Source: Wilson (1995).

An Empirical View of Loss Severity

In theory, we would like to use the decomposition presented above to come up with a simple model for loss severity. Specifically, given a defaulted loan, we would compute the following estimates:

- **Base severity.** The sale price for the foreclosed property could be estimated using a home price index for the region that the loan is originated in.
- **Uncollected interest.** This is equal to the number of missed interest payments between the time of the last timely monthly payment and the time the house was liquidated.
- **Average levels for various expenses.** Realtor commissions, foreclosure costs, and holding costs.

When summed together, these different components should give us an estimate of the expected loss severity. Unfortunately, the loan level data sets that we have access to report only total loss and not the different components of loss. Also, there are a number of measurement issues associated with the calculation of base severity. In short, the decomposition of loss severity presented in Figure 6.2 is mostly useful in a conceptual sense. It also ignores other factors that can impact loss severity such as state effects (see Figure 4.1 for an illustration of how foreclosure expenses vary by state), servicer effects (some servicers are more efficient in terms

³¹ See Donald Wilson, *Residential Loss Severity in California: 1992 to 1995*, Journal of Fixed Income, December 1995.

³² The analysis did not cover recoveries from private mortgage insurance.

of resolving defaults than others), borrower characteristics (loan sizes and loan-to-value ratios can impact loss severity), loan purpose (refinance loans tend to be associated with higher loss severities), occupancy type (loss severities are higher on investor properties) etc. The takeaway is that in practice we forecast loss severity by constructing and estimating a multivariate regression model with explanatory variables that serve as proxy variables for the different factors influencing loss severity as per our discussion.

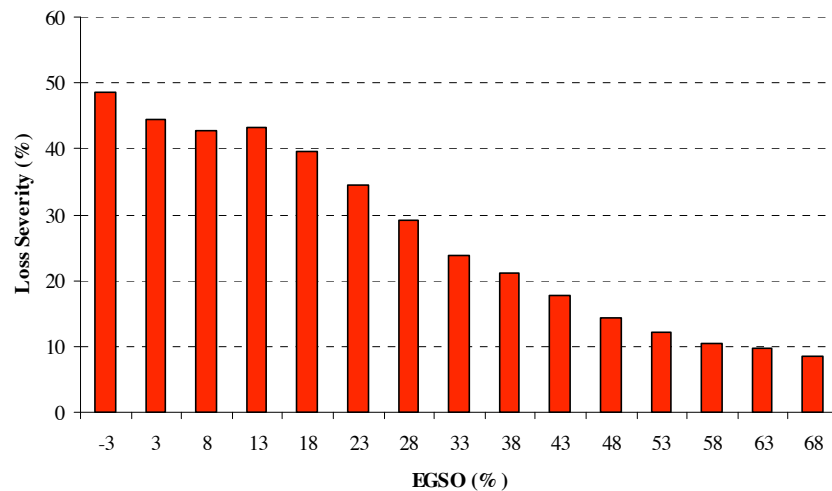
Still, to move from the realm of the conceptual to the practical, it is useful to get some sense of the typical magnitude of loss severity on a residential mortgage and how these numbers vary as a function of the different variables we have discussed above. The goal here is not to provide an exhaustive catalog of all the different factors that can determine loss severity but instead provide pictures of some of the important relationships. The pictures are one-dimensional (How does Loss Severity depend upon Loan Size?), whereas in reality loss severity will be depend upon the interplay of several factors with some factors increasing loss severities and others offsetting it. The data we show on loss severity corresponds to loss data on fixed-rate 30-year subprime loans with an age greater than 20 months originated between 1990 and 2007. There is nothing special about our choice of 30-year fixed-rate subprime loans: the relationships we discuss below would also hold for other mortgage product types (ARMs, for example) and different sub-sectors of the RMBS market (Conforming, Alt-A, Jumbo).

Base Severity

Base severity is the largest component of loss severity and consequently a crucial part of the loss severity calculation. While zip-code or MSA level home price indices from OFHEO or Case-Shiller can be used to estimate the value of the foreclosed home, it is important to realize that the HPIs in question give us home price appreciation for the *average* home. Appreciation on a foreclosed property is likely to be much less than the average property which means that the HPI-implied house value needs to be appropriately haircut.

Another nuance to consider is that the price of the foreclosed home relative to the average property (the haircut) may be a function of market conditions. In fact, the Wilson paper appears to suggest that in a good market, foreclosed properties may trade at a greater discount to the average property than in a soft market. The argument being that a greater degree of adverse selection must be present for a property to default in a good market.

Figure 6.3 provides a perspective on some of these issues by plotting the loss severity experienced by loans in our sample as a function of the average equity growth they have witnessed since origination (ESGO). The figure shows that loss severities range from less than 10% to nearly 50% as a function of housing market conditions. Notice that even in an environment where the average home had experienced 60% equity growth, loans do default, although the loss severity is less than 10%. Similarly, on the other side of the distribution, in an environment where the average loan experiences ESGO of 3%, average loss severity can average nearly 50%. The housing market downturn of 2007-2008 will give us a sense for what loss severities will average in more extreme scenarios of home price declines.

Figure 6.3: Loss Severity on Fixed-Rate 30-year Subprime Mortgages by EGSO

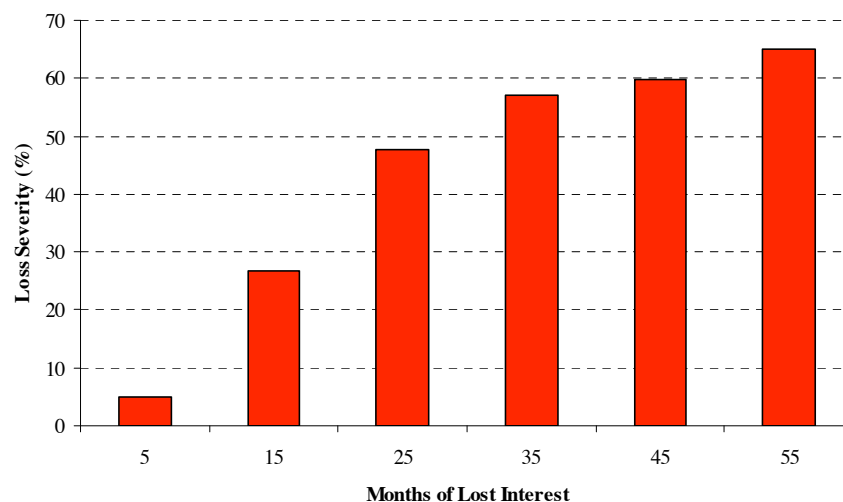
Source: Banc of America Securities LLC estimates

Uncollected Interest

The relationship between loss severity and lost interest is intuitive and is given by:

$$\text{Uncollected Interest} = \text{Loan Coupon} * \text{UPB} * \text{Period between time of default and property disposition}$$

Figure 6.4 captures this dependence in a simple way by simply plotting loss severity as a function of the number of missed payments by the homeowner.

Figure 6.4: Loss Severity on Fixed-Rate 30-year Subprime Mortgages by Lost Interest

Source: Banc of America Securities LLC estimates

Expenses

There are several expenses associated with the foreclosure process: realtor commissions, foreclosure costs, and holding costs. While realtor commissions are straightforward to understand, there is more complexity to foreclosure costs and holding costs.

Foreclosure costs vary by state since there is state-level variation in foreclosure laws. As per our discussion in Section IV, judicial foreclosures are more expensive than non-judicial foreclosures because they require lawyers as opposed to trustees and can take one to two years to complete, as opposed to an average of 4 months for a non-judicial case. Adding to the expense of judicial foreclosures is the fact that some states permit a right of redemption after a judicial foreclosure. During the one year redemption period, the lender still has to pay property taxes, hazard insurance etc. Based on the studies we have seen, when aggregated across states, foreclosure expenses tend to average between 5% to 8% of the UPB.

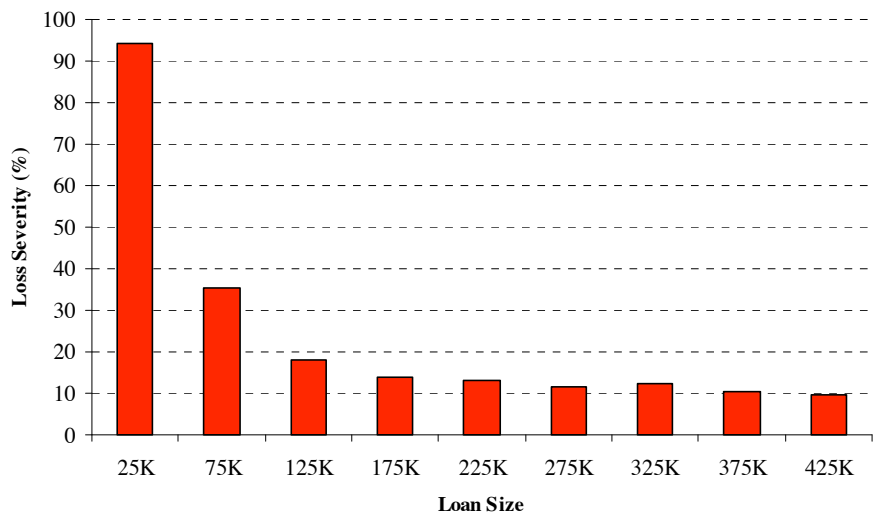
In the situation that a lender needs to hold onto an REO property, there's an additional layer of costs that include property maintenance costs, utilities, property taxes and property insurance. These costs are time-dependent in that they will be proportional to the amount of time between obtaining title to the property and disposing of it. This time period averages about 7 months but can vary significantly depending upon market conditions.

Other Factors Influencing Loss Severity

There are several other factors that influence loss severity and we provide an overview of some of them in Figures 6.5 through 6.8. For the sake of brevity, our overview is not comprehensive. For example, loss severity can also depend upon the state of origination (as Figure 4.1 shows) and the servicer. Some collateral characteristics that play a key role in determining loss severity are:

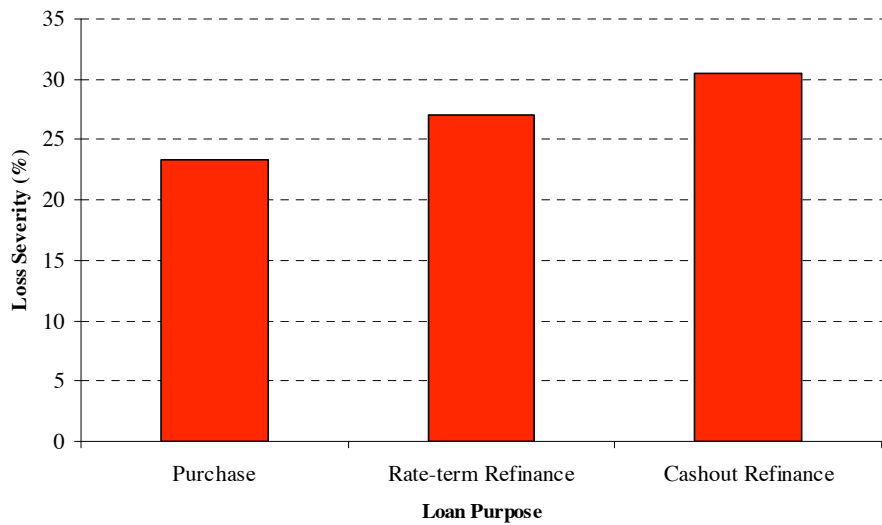
- **Loan Size.** Lower loan sizes are associated with higher loss severities because of the fixed costs of the foreclosure process along with the fact that servicers have more of an incentive to minimize loss severity on larger loan balances since the absolute dollar amount of loss can be large even at small severities. See Figure 6.5.
- **Loan Purpose.** Loss severities are higher on refinance loans as opposed to purchase loans (after controlling for other characteristics). Studies suggest that this may be a result of appraisal values being biased upwards in refinance transactions. (However, recent data from the 2006 vintage are showing higher loss severities for purchase mortgages.)
- **Occupancy Status.** Loss severity is greater for investor properties. Investor properties are more illiquid than owner-occupied properties which may result in higher base severities. Also, investors may be more ruthless in terms of exercising their default option and walk away from the property when it is clear that any recovery value would be minimal.
- **Original LTV (OLTV).** There is a pronounced relationship between OLTVs and loss severities with loss severity increasing at higher OLTVs. Given borrowers that have experienced the same amount of equity growth (as captured by an HPI), the borrowers with higher OLTVs will clearly be associated with higher loss severities, all else being equal.

Figure 6.5: Loss Severity on Fixed-Rate 30-year Subprime Mortgages by Loan Size



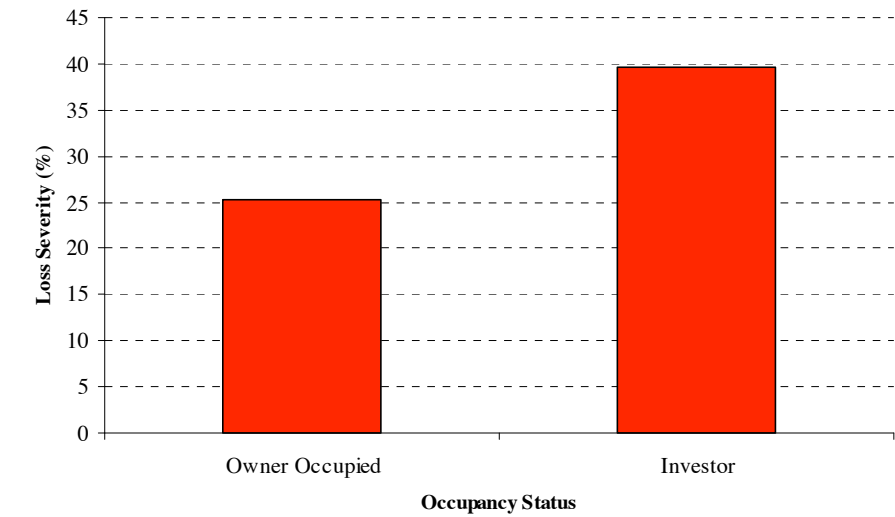
Source: Banc of America Securities LLC estimates

Figure 6.6: Loss Severity on Fixed-Rate 30-year Subprime Mortgages by Loan Purpose



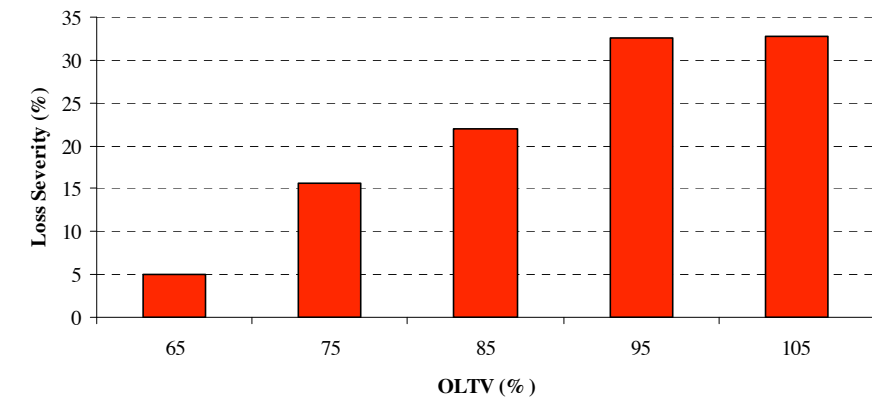
Source: Banc of America Securities LLC estimates

Figure 6.7: Loss Severity on Fixed-Rate 30-year Subprime Mortgages by Occupancy Status



Source: Banc of America Securities LLC estimates

Figure 6.8: Loss Severity on Fixed-Rate 30-year Subprime Mortgages by OLTV



Source: Banc of America Securities LLC estimates

Loss Severity By RMBS Sector and Mortgage Product Type

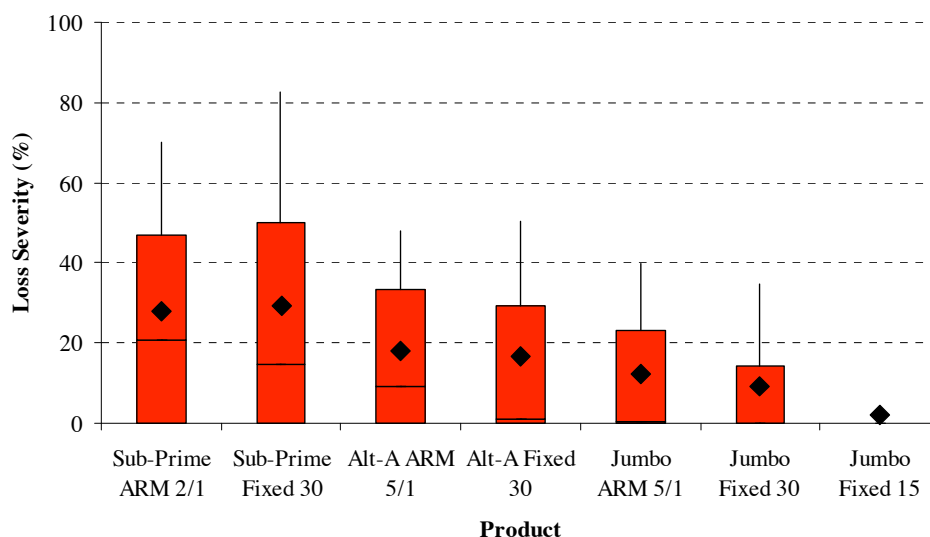
The relationship between loss severity and the factors that we have considered do not just hold for subprime fixed-rate loans. The same relationships should hold (they may be weaker or stronger) for other subprime mortgage product types (2/28 ARMs, 40-year mortgages) and across different sectors of the residential mortgage sector (alt-A fixed-rate mortgages, alt-A

ARMs, jumbo 30-year mortgages etc.). This type of data is best shown through a box-and-whisker plot (see Figure 6.9) which is a convenient way of summarizing and comparing the loan-level distribution of loss severities:

- The ends of the box correspond to the highest value in the lower (Q1) and upper quartile (Q3).
- The median value is marked by a horizontal line inside the box, whereas the weighted-average value is denoted with a diamond.
- The whiskers are the two lines outside the box that extend to the maximum and minimum value.
- Loss severity is bounded below at 0% and the figure shows that there are a number of defaults with this severity rate since the lower ends of all of the boxes are flush with the 0% rate (which also explains why there is no lower whisker).

As we can see from the figure, loss severities on subprime collateral are in general higher and more widely dispersed than for prime mortgage products. Loss severities on the prime products typically average in the mid-teens for fixed-rate 30-year product and are much lower for 15-year product.

Figure 6.9: Loss Severity by RMBS Sector and Mortgage Product Type*



*For loans originated between 2000-2007.

Source: Banc of America Securities LLC estimates

APPENDIX A. MBA AND OTS DELINQUENCY CALCULATIONS FOR LOANS

There are two key dates to keep in mind in delinquency calculations: the **due date** and the **cut-off date**. The due date on most mortgages is the first day of the month³³, while the cut-off date is typically the end of a calendar month. The **delinquency status** calculation places a loan in the appropriate delinquency category (30-day Delinquent, 60-day Delinquent, 90-day Delinquent, Foreclosure) based on one of two prevailing conventions – OTS and MBA. The Office of Thrift Supervision (OTS) calculation is typically used for home equity/subprime products while the Mortgage Banker's Association (MBA) calculation is used for traditional prime products. Depending on the due date of the loan, the two calculation methods can give a delinquency status that differs by one payment (i.e., 30 days).

In the OTS calculation, a loan is considered delinquent if a monthly payment has not been received by the close of business on the loan's due date *in the following month* (i.e., by the next due date). For example, a loan with a due date of May 1st with no payment received by the close of business on May 31st (the cut-off date in this example) would be reported as current in June statements. As long as the payment is made by June 1st, the borrower is not considered delinquent in the OTS calculation. Assuming no payments were made during June, the loan would be reported as 30-days delinquent in the July statement.

In the MBA calculation, a loan is considered delinquent if the payment has not been received by the end of the day immediately preceding the loan's next due date. Generally this means that the payment should be received by the last day of the month the payment was due. Therefore, the borrower with a due date of May 1st who had not made their payment by May 31st would be reported as 30-days delinquent in the June statement. Assuming no payments are made during June, the loan would be reported as 60-days delinquent in the July statement. Figure A.1 illustrates the different answers we get under the two different methods for a mortgage payment that is made at various different times.

Figure A.1: Comparing the MBA and OTS Delinquency Calculations

Due Date	Next Due Date	Cut-off Date	Payment Made	Loan Status	
				MBA	OTS
May 1	June 1	May 31	May 31	Current	Current
			June 1	30-day	Current
			July 1	60-day	30-day

Source: Banc of America Securities LLC.

When the due date on a loan falls on any other day besides the first of the month, the MBA and OTS calculations give the same answer, assuming an end of calendar month cut-off date. For example, if a mortgage payment is due on May 2nd and has not been received by the cut-off date of May 31st, both the MBA and OTS methods will record the loan as current. For the OTS to record the loan as delinquent, the mortgage payment would not have been received by June 2nd, while the MBA would record the loan as delinquent if the payment were not received by June 1st. Finally, foreclosures and REOs are treated identically by both methods.

³³ For example, the April payment on a mortgage is due on May 1st.

APPENDIX B. SEASONAL FACTORS IN MORTGAGE DELINQUENCY

Mortgage delinquency shows some distinct seasonal patterns which are summarized in Figure B.1. Briefly, there's a noticeable increase in delinquency rates at the end of the year and an equally pronounced decline over the first quarter of the year. What explains these seasonal effects in mortgage delinquency? Section II suggested two likely reasons: (1) Incomes for workers in seasonal industries such as housing, construction or agriculture may fluctuate in sympathy with the seasonal trend in activity in these sectors and (2) Families may tend to over-extend themselves over the holiday season and then play catch-up after Christmas.

Figure B.1: Estimated Seasonal Factors for Residential Mortgage Delinquency

<u>Month</u>	<u>Seasonal Factor</u>
January	1.02
February	0.95
March	0.90
April	0.91
May	0.92
June	0.96
July	0.99
August	0.95
September	1.08
October	1.09
November	1.15
December	1.21

Source: Banc of America Securities LLC.

What allows families to catch up on their finances? The timing of year-end bonus payments and, perhaps more importantly, the timing of income tax refunds. Surveys of individuals typically show that most of them (60%-85%) either save the tax refund, use it to pay bills, or pay down debt. Take a look at Figure B.2 which shows the distribution of the timing of income tax refunds and is generally supportive of the seasonal pattern in that most of the refunds are concentrated in the months of February-April, the months in which the seasonal factors for delinquency are among the lowest.

Figure B.2: Timing of Individual Income Tax Refunds: 2004-2007

Month	% of Annual Refund Disbursements
January	3%
February	31%
March	25%
April	22%
May	11%
June	2%
July	2%
August	1%
September	2%
October	1%
November	3%
December	1%

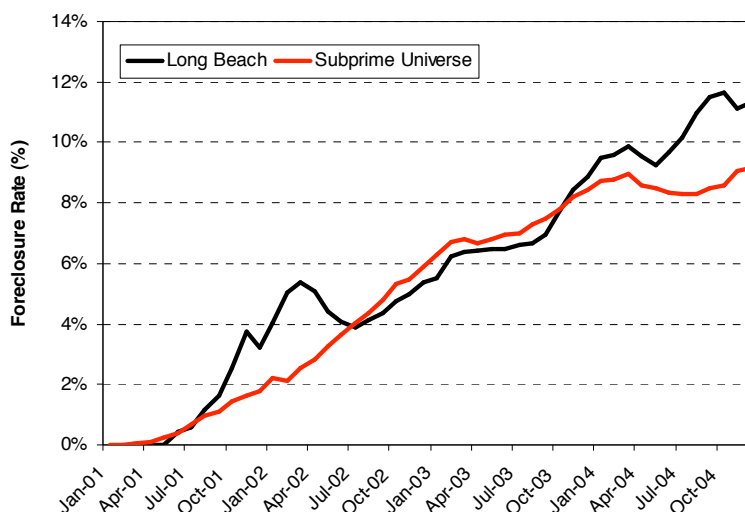
Source: U.S. Department of the Treasury, *Daily Treasury Statements*, various issues.

APPENDIX C. MONITORING SERVICER PERFORMANCE

The Impact of Servicing on Collateral Performance

The dependence of mortgage credit performance on proper servicing cannot be overemphasized. In particular, subprime borrowers require more attentive and aggressive servicing than prime borrowers. For example, let's walk through an example related to the servicing of Long Beach deals in 2001. Figure C.1 shows foreclosure rates on 2001 vintage Long Beach collateral versus the 2001 vintage home equity universe. Notice the significant jump in foreclosure rates for Long Beach collateral from 3Q 2001 through 2Q 2002. This was a direct result of deterioration in servicing quality. After Washington Mutual (WAMU) acquired Long Beach Mortgage Corporation in 4Q 1999, the Orange County based subprime servicing operation was consolidated with the WAMU's core prime servicing operations. Many key associates were lost in the transfer leaving the subprime servicing platform deficient in expertise. Consequently, the servicing techniques employed proved ineffective on subprime borrowers and resulted in severe collateral underperformance. As a result, WAMU revamped its subprime servicing in Northridge, California and installed a senior management team with extensive subprime servicing experience. The new management team emphasized employee training and hired experienced personnel from the subprime servicing industry. Following these initiatives, collateral performance improved dramatically with foreclosure rates returning to industry averages by 3Q 2002.

Figure C.1: 2001 Vintage Foreclosure Rates of Long Beach and the Home Equity Universe



Source: Intex, Banc of America Securities

Monitoring Servicer Performance

Given that proper servicing of mortgages is critical to collateral performance, investors should closely monitor servicers to ensure collateral performance falls within expectations. Three of the key ways in which investors monitor servicer performance are through rating agency reports and ratings, delinquency trend analysis, and roll rate analysis.

All three rating agencies publish periodic reports and ratings of mortgage servicers, including

ratings specific to subprime primary servicing. Figure C.2 shows subprime primary servicer ratings (as of May 2007) for the ten largest subprime servicers. The Moody's rating scale ranges from SQ1 (strong) through SQ5 (weak). S&P rates servicers as STRONG, ABOVE AVERAGE, AVERAGE, BELOW AVERAGE, or WEAK. In addition, S&P assigns a near-term outlook of Positive, Stable, Developing, or Negative. Fitch assigns ratings of RPS1 (full approval) through RPS5 (conditional approval). Within the rating levels, Fitch differentiates the rankings by pluses (+) and minuses (-).³⁴

Figure C.2: Subprime Servicing Ratings as of May 2007

Servicer	Moody's	S&P	Fitch
Countrywide Financial	SQ1	Strong	RPS1
Chase Home Finance	SQ1	Strong	RPS1
Option One Mortgage Corp.	SQ2+	Strong	RPS1
Ameriquest Mortgage Corp.	SQ2+	Strong	RPS3+
CitiFinancial (E)	SQ2	Above Average	RPS1-
Ocwen Financial Corp.	SQ2-	Strong	RPS2
Wells Fargo Home Mtg.	SQ1	Strong	RPS1
Homecomings (E)	SQ2+	Strong	RPS1
National City Home Loan	SQ1-	Above Average	RPS2
HSBC Mortgage Services		Strong	RPS2-

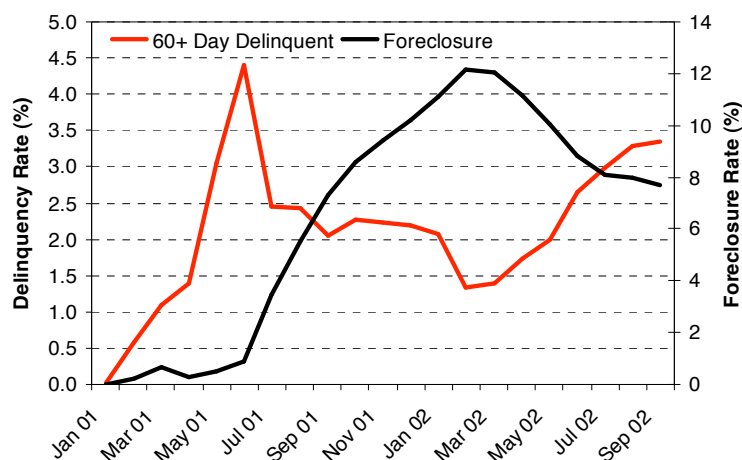
Source: Banc of America Securities

The rating agencies adjust credit enhancement requirements in securitizations based on these ratings. They decrease enhancement requirements in securitizations with higher-rated servicers and increase enhancement requirements for lower-rated servicers. When determining the rating, the agencies stress collection management and default management capability. However, there are many less quantifiable attributes that are also considered in the rating process. These include M&A, management experience, financial condition, staffing and training, employee turnover, operational controls and procedures, and technological sophistication. Such attributes will affect the operational and financial stability of the servicer and impact the servicer's ability to adapt to changing market conditions. For example, on April 25th 2007, Moody's downgraded Option One's subprime primary servicer rating from SQ1 (strong) to SQ2+ (above average) based purely on an assessment of reduced servicing stability to average from above average, in response to Option One's sale to Cerberus Capital Management, L.P. This is despite Moody's view that Option One's collection abilities are "strong" and acknowledgement that Option One had made various enhancements to its servicing platform.

As rating agencies only periodically update servicer ratings and reports, delinquency trend analysis can be used to identify servicing quality deterioration in a more timely manner. Since delinquencies are often a precursor to eventual default, unusual and steep increases in delinquency rates may reflect servicing quality deterioration that will result in higher collateral losses. We again examine the servicing difficulties experienced in Long Beach collateral beginning in 2001. Figure C.3 shows 60+ day delinquency rates and foreclosure rates for Long Beach (LBMLT) 2000-1. Notice the sharp spike in delinquencies peaking in July 2001 that foreshadowed the surge in the foreclosures.

³⁴ The overall servicer rating is based on several criteria. For example, Fitch considers takes the following criteria into account in order to reach a final rating: company and management experience, financial condition, staffing and training, procedures and controls, loan administration, default management, technology, and oversight and monitoring of subservicers. See *Rating U.S. Residential Mortgage Servicers*, Fitch Ratings, November 29th, 2006 for further details.

Figure C.3: Long Beach 2000-1 60+ Delinquency Rates and Foreclosure Rates



Source: Intex, Banc of America Securities

Deeper insights into servicer performance and strategies can be obtained through roll rate analysis. For example, Figure C.4 shows average monthly roll rates for Long Beach 2000-1 collateral from 2001 to 2002. We can infer that, on average, 4% of current loans migrated to 31-60 day delinquency status each month. Conversely, 35% of 31 to 60 day delinquent loans returned to current status. Roll rates can also be used to monitor the effectiveness of collection management and default management. Lower roll rates from current status to delinquent status would show superior collection management. Likewise, higher roll rates from foreclosure or 91+ day delinquency statuses to current status would indicate application of loss mitigation techniques.

Figure C.4: Average Monthly Roll Rates for Long Beach 2000-1 from 2001 through 2002

From Status	To Status						
	Current	31-60 days deliq	61-90 days deliq	91+ days deliq	Foreclosure	REO	Paid Off
Current	93%	4%	0%	0%	0%	0%	2%
31-60 days deliq	35%	27%	21%	1%	12%	0%	3%
61-90 days deliq	10%	12%	19%	40%	16%	0%	2%
91+ days deliq	1%	1%	2%	79%	15%	1%	1%
Foreclosure	3%	1%	0%	6%	83%	4%	3%
REO	0%	0%	0%	1%	1%	90%	8%

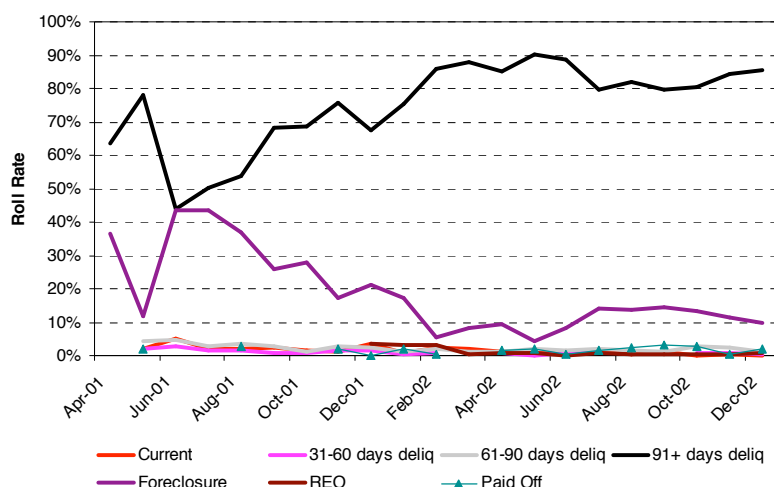
Source: Loan Performance, Banc of America Securities

Monitoring changes in roll rates over time can be useful in analyzing the evolution of servicing procedures and capability. For example, an increase in monthly roll rates from current status to delinquent status could indicate deterioration in servicing effectiveness. In addition, changes in the transitions of seriously delinquent loans could indicate a strategic shift in which types of loss mitigation techniques are being utilized. Figure C.5 shows monthly roll rates from the 91+ days delinquent state to various other states for Long Beach 2000-1 from April 2001 to December 2002. Notice how the roll rates from 91+ days to foreclosure change over time. In June 2001, over 40% of 91+ day delinquent loans migrated to foreclosure status in July. However, transitions to foreclosure fell to under 10% in February 2002. Such dramatic changes

³⁵ The consent of the trustee may be needed in this case.

in roll rates on the same portfolio reflect fundamental shifts in servicing procedures.

Figure C.5: Changes in Roll Rates for Long Beach 2000-1 from 91+ Delinquent State to Various Other States Over Time



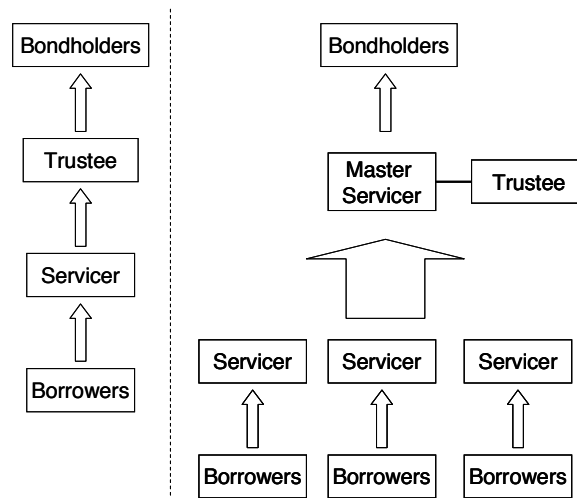
Source: Loan Performance, Banc of America Securities

Servicing Transfers

Monitoring servicing becomes particularly important in the context of servicing transfers, so we will spend some time going through the details associated with a transfer of servicing rights. In a nutshell, a servicing transfer is the assumption of the servicing rights and obligations by a different corporate entity. Servicing transfers can happen for many reasons. They may take place as part of the regular course of business due to the sale of servicing rights, or because of mergers/acquisitions. Transfers may also happen due to irregular events such as a servicer event of default as per the pooling and servicing agreement (PSA) corresponding to a particular trust, or a servicer resignation.³⁵ Clearly, the focus of monitoring here is to determine to what extent the new servicer's business practices are different from the previous one's, and to assess the potential impact of this change on performance.

A Brief Overview of Servicing and Servicing Contracts

Let's quickly review how mortgage payments flow from borrowers to bond holders in a typical RMBS transaction. Figure C.6 shows two scenarios based on the number of servicers involved in a particular transaction. At one extreme is a very simple setup where all of the loans in a deal are serviced by one servicer. The servicer takes in all monthly collections, which include mortgagors' monthly payments (or advances by the servicer to the extent payments have not been received from borrowers), liquidation proceeds, prepayments and prepayment premiums, and compensating interest. These collections are passed on to the trustee, who in turn distributes them to bondholders. At the other extreme, we have a situation where the loans in the deal are serviced by multiple servicers. These deals typically have a **master servicer**, whose job is to oversee the individual servicers, pool the monies collected or advanced by each servicer, and forward monthly collections to bond investors.

Figure C.6: Flow of Mortgage Payments from the Borrowers to the Bondholders

Source: Banc of America Securities

Parties to the PSA typically include the trustee, the depositor/seller, and the master servicer. If there is no master servicer, then the underlying servicer(s) are generally party to the PSA. The servicing fee for the loan is retained by the servicer and is generally set at a level that is required by the rating agencies. Current fees average 50bps for subprime and second lien collateral, 37.5 bps for option ARMs and certain hybrid ARMs, and 25 bps for most other collateral. The servicing fee is generally more than sufficient to cover servicing expenses and, as a result, servicers typically make an upfront payment to obtain mortgage servicing rights. The rationale for setting servicing fees at a higher level than expenses is to avoid any problems in obtaining servicers for seasoned mortgage portfolios which are typically associated with higher servicing expenses.

Servicing Transfers Resulting from Servicer Default

Understanding servicing transfers because of servicer default is a central issue for many investors so we go through some of the details associated with this situation. The PSA lays out a set of terms and conditions that the servicer is expected to comply with and non-compliance with these requirements may result in a default event. Unless the servicer cures the default within a certain time frame (depending on the severity of the lapse), the servicing contract may be terminated by the trust. In some cases, the servicer may get an extension to cure the violation or obtain a waiver for an event of default if more than a certain percentage of certificate holders agree.

Some of the common violations that would result in an event of default include:

- Failure to remit required payments (other than P&I advances from their own funds) to the securities administrator (~1-2 days to cure).
- Failure to make the required P&I advances from their own funds on the remittance date. The servicer usually has until the next day to cure the default.
- A voluntary bankruptcy filing.

- An involuntary bankruptcy filing that remains undischarged for a certain period (~90 days).
- The appointment of a conservator, receiver, or liquidator.
- Material breach of any representation and warranty (~30-60 days to cure).
- Failure to perform or observe any covenant or agreement in the PSA (~30-60 days to cure).
- Failure to maintain a certain servicer rating.
- Exceeding certain loss/delinquency triggers.
- Ceasing to be an approved Freddie Mac/ Fannie Mae seller/servicer.

Successor Servicers for Servicing Transfers

When a servicer merges with or is acquired by another corporation, the new company simply takes over the servicing. The surviving corporation may be required to meet certain conditions like being qualified to sell mortgage loans to and being able to service mortgage loans for Freddie Mac or Fannie Mae. If servicing rights are simply sold in the normal course of business, the entity that buys the servicing rights takes over the servicing. In general, consent from the trustee or the master servicer and a letter from each rating agency stating that the servicing transfer will not impact the current rating of the certificates are also needed. Furthermore, the successor in this scenario may also need to meet certain additional criteria like Fannie Mae and Freddie Mac eligibility or a minimum servicer rating. In the situation of a servicer resignation, the successor is identified either by the outgoing servicer or the trustee/master servicer. A servicer's resignation is not effective until the successor servicer has assumed the outgoing servicer's liabilities and obligations under the PSA. Just like in the previous scenario, consent is needed from the master servicer or the trustee, certain eligibility requirements have to be met, and a "no downgrade letter" from each rating agency may be needed.

In the case of a servicer default, the trustee/master servicer/depositor (depending on the terms of the PSA) is typically required to terminate servicer obligations and rights. The servicing is then fully transferred to the one of the following entities – the backup servicer, the master servicer, the trustee, or a servicer appointed by the master servicer/trustee, depending on the PSA. PSAs generally specify a time frame within which servicing must be transferred to the successor and this period typically ranges from 2-3 months.

The Mechanics of Servicing Transfers

Servicing by its very nature is fairly tactical and transferring servicing without any disruption requires coordination on multiple fronts. First and foremost, borrowers need to be notified about redirecting their mortgage payments to the new servicer. Despite the apparent simplicity of this task, sloppy execution of logistics may result in artificially higher delinquencies in the short term. In general, the outgoing servicer is required under consumer protection laws to notify the borrower of the transfer at least 15 days before the **effective date**³⁶ of the transfer. The successor servicer is also required to notify the borrower within 15

³⁶ The effective date is the date when the first mortgage payment is due at the new servicer's address.

days after the transfer has occurred. In mortgage industry parlance, these borrower notifications are also called “hello” and “goodbye” letters. There is a 60-day grace period after the transfer date during which no late payment fee can be charged if timely payments were made to the outgoing servicer.

As part of the servicing handoff, the outgoing servicer transfers a copy of the underlying mortgage note to the successor servicer. This is needed so that the new servicer can enforce the terms of the mortgage note. In addition, the outgoing servicer transfers to the new servicer loan level servicing files that contain payment histories for the loans. Usually, the servicing files are in electronic format and if the servicing systems for the outgoing servicer and the incoming servicer are not the same, then additional effort will be needed during the “on-boarding” process. For effective servicing transfers, it is of paramount importance that all these steps are completed as soon as possible. Delays in servicing transfer are much more likely to impact the performance of weaker credit borrowers than prime quality borrowers.

The speed with which servicing can be transferred also depends on the role of the successor servicer prior to the transfer. In cases where the successor servicer also happens to be the backup servicer who is providing a “hot” backup, the servicing can be transferred almost immediately. This is because a servicer who acts as “hot” backup essentially maintains duplicate servicing systems and processes. In contrast, a servicer acting as a “cold” backup gets involved only after a servicer event of default has materialized. Generally, servicing transfers that involve “cold” backups do not appear to be materially different from transfers where no backup was assigned.

If the servicing transfer is triggered due to servicer default, the defaulting servicer must fulfill certain legal obligations in order to smoothen the transition. In particular it is required to:

- Cooperate in the servicing transfer;
- Provide all servicing files and documents in a timely manner;
- Promptly transfer all the collected payments into the trust account; and,
- Pay for expenses related to the termination (servicing transfer costs and legal expenses).

In addition, PSA documents generally require that the successor servicer not receive a servicing fee that is in excess of what the outgoing servicer collected. Unless the performance of the mortgage portfolio has deteriorated significantly this requirement is usually not a problem because servicing fees are typically high enough to cover the additional expenses associated with servicing a seasoned portfolio.

Special Situation: Servicer Bankruptcy

There are several important issues that need to be kept in mind if a servicing transfer is triggered due to servicer bankruptcy.³⁷ As we mentioned earlier, in general servicing is a profitable business and a servicer has to make an upfront payment to acquire servicing rights. When the parent company of the servicer files for bankruptcy, both the existing servicer and the debtors of the parent company will want to realize the value of their servicing assets for their own benefit. The problem is that once there is an event of default, deal documents

³⁷ The Fitch report “*U.S. Residential Mortgage Servicer Bankruptcies, Defaults, Terminations, and Transfers*,” discusses these issues related to servicer bankruptcies in detail.

generally provide that the master servicer/trustee can take over the servicing without *any* compensation to the existing servicer or debtors, thus effectively valuing this asset at zero from the perspective of these two parties. To avoid immediate termination of their servicing rights, the servicer and the debtors may utilize certain protections available to businesses under the U.S. Bankruptcy Code. First of all, the Code provides for a “breathing spell” during which businesses may attempt to reorganize or put themselves up for sale as a going concern. During this period they are protected against any remedial action from creditors and other contract partners. Furthermore, servicers may use protection offered under the “automatic stay” provision of the U.S. Bankruptcy Code, which prohibits creditors and contract partners from collecting a debt, terminating a contract, or exercising control over the property of the estate. Thus, servicers get automatic protection against termination of the servicing contract for a short period of time. This, however, does not provide long term relief for the servicers.

For longer term relief, servicers may argue that PSA is an **executory contract**³⁸ and may again seek protection under the U.S. Bankruptcy Code. If the court agrees, the servicer is not only given an opportunity to pick and choose which contracts it wishes to assume and which ones it wants to dispose of, but is also allowed to take its time to decide, which is until the end of the bankruptcy case. If the bankrupt servicer manages to assume the contract then it continues servicing the deal. However, if there are any other contractual breaches by the servicer that remain uncured or if bondholders get negatively impacted due to the bankruptcy protection proceedings, the trustee may seek relief and may be able to have the servicing contract terminated. Alternatively, the trustee may sometimes be able to appeal the “automatic stay” by showing that the servicer is unlikely to be able to reorganize the business or sell servicing rights for profit. In general, the best way for the trustee to proceed is to proactively find a solution even before the bankruptcy petition is filed.

Case Studies of Servicing Transfer

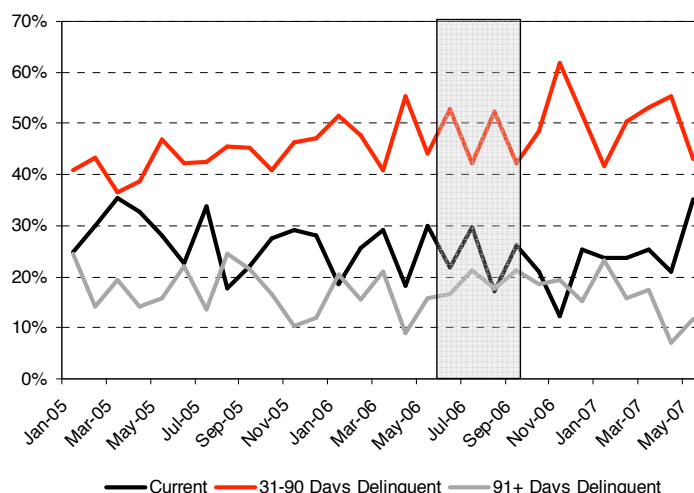
We now present a few case studies to highlight some of the issues related to servicing transfer. The roll-rate metrics we had presented earlier in the section constitute an effective way of examining both the short term and long term impact of servicing transfer. For our analysis, we consider roll rates from the 31-90 days delinquency state to the following three states - Current, 31-90 days delinquency (combination of 31-60 days and 61-90 days delinquent states), and 91+ days delinquency. The reason is that the roll rates from early stage delinquency are likely to be most sensitive to servicing disruptions during the transfer or to changes in servicing methodology. Also, for our analysis we focus only on servicing transfer associated with portfolios of ARM loans.

Case Study 1: Sale of U.S. Home Mortgage Servicing Business of HomeQ to Barclays Announced on 6/22/2006

This transaction was more of a corporate acquisition where the whole servicing business unit was transferred to Barclays with very little impact at the operational level. As a result, there were no real issues involved with the servicing transfer. This can be clearly seen in Figure C.7, which shows that roll rates from 31-90 days delinquency state to various other states did not change in any meaningful way after the transfer.

³⁸ A contract in which both parties have some obligations that they still need to perform.

Figure C.7: Roll Rates from Early Stage Delinquency to Other Stages for Loans Serviced by HomeQ (Sale to Barclays announced on 6/22/2006)

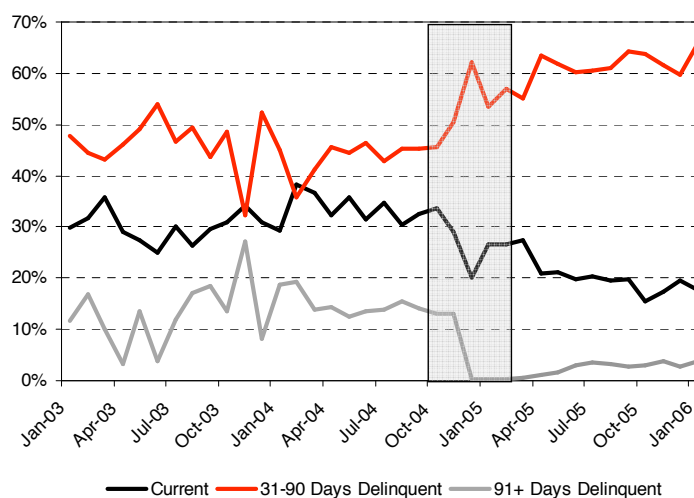


Source: Loan Performance, Banc of America Securities

Case Study 2: Sale of Several Assets of PCFS Mortgage Resources (Division of Provident Bank) to C-Bass on 10/08/2004

In this acquisition it appears that even though there were no significant short term disruptions, servicing practices changed after the transfer. This can be observed in Figure C.8 which shows that rolling delinquencies went up over time (roll rates to 31-90 days delinquency state went up) and roll rates to 91+ Days delinquency dropped. Essentially, the new servicer was able to make delinquent borrowers resume making timely payments. This is a positive development both from the perspective of servicers and investors and demonstrates superior loss mitigation practices by the new servicer.

Figure C.8: Roll Rates from Early Stage Delinquency to Other Stages for Loans Serviced by Provident (Sale to C-Bass announced on 10/08/2004)

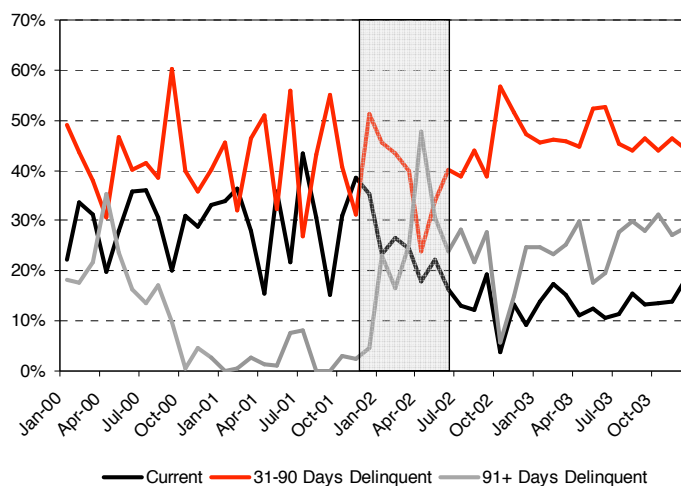


Source: Loan Performance, Banc of America Securities

Case Study 3: Sale of EquiCredit to Fairbanks Announced on 12/07/2001

In this transaction it seems that there were some short term disruptions in servicing as indicated by spike in roll rates to 91+ days delinquency state and a simultaneous drop in roll rates to 31-90 days delinquency state (Figure C.9).

Figure C.9: Roll Rates from Early Stage Delinquency to Other Stages for Loans Serviced by EquiCredit (Sale to Fairbanks announced on 12/07/2001)



Source: Loan Performance, Banc of America Securities

APPENDIX D. STATE-LEVEL VARIATION IN FORECLOSURE LAW³⁹

State-level variation in foreclosure law principally occurs along the following three dimensions:

- The foreclosure procedure followed: Judicial, power-of-sale, or other forms of foreclosure;
- The availability of deficiency judgments; and,
- The availability and duration of the statutory redemption period.

Foreclosure Process

The key differences between power-of-sale and judicial foreclosure have already been explored in a fair amount of detail in Section IV. Figure D.1 displays all the states that require a judicial foreclosure process. The unshaded states thus correspond to states where both power-of-sale and judicial foreclosure are available.

Deficiency Judgments

In practice, whether the lender can sue the borrower to recover the difference between the outstanding mortgage debt and the foreclosure sale price plus default-related costs (the **deficiency amount**) varies by State. As Figure D.2 shows, approximately 9 states effectively prohibit deficiency for the typical mortgage default case (single-family, owner-occupied homes). The point of these statutes is to discourage lenders from “back-bidding” at the foreclosure sale thus increasing the amount of the deficiency along with potentially being able to flip the property for a quick profit. We say “effectively prohibit” above because, apart from Rhode Island, no States explicitly restrict deficiency judgments. Instead, they make it very difficult for a lender to pursue this course of action by making it very expensive to do so. We discuss the case of California which is fairly representative. In general, no deficiency judgments are allowed in this State unless the lender does a judicial foreclosure or if the foreclosure concerns a purchase mortgage. However, a lender will rarely choose to opt for a judicial foreclosure:

- They are more expensive than non-judicial foreclosures since they entail hiring a lawyer as opposed to non-judicial foreclosures which are done by private trustee companies for a flat fee.
- Judicial foreclosures last much longer. A non-judicial foreclosure can be done in about 4 months in California while judicial foreclosures can last over a year and can even extend to two years if contested.
- The homeowner has a **right to redemption** in a judicial foreclosure in this state. This right provides the borrower with the ability to buy back their mortgage after the foreclosure sale within a limited period of time (the **redemption period**). This redemption period lasts for one year in California unless the lender makes a full price bid in which case the period is shortened to three months. If the mortgagor is able to buy back the house for the foreclosure sale price plus expenses, they are allowed to

³⁹ Our discussion is based on Karen M. Pence, *Foreclosing on Opportunity: State Laws and Mortgage Credit*, Federal Reserve Board, May 2003.

keep the property. The homeowner is allowed to stay in the house over the redemption period. If they are unable to redeem the property by the end of the period, the purchaser at the foreclosure sale receives title to and possession of the property.

In general, even in “pro-deficiency” States, deficiency judgments are rarely used except to negotiate with the borrower.⁴⁰ They are more widely used in situations involving fraud or non-hardship. For example, a lender may be more likely to sue an investor or a borrower who is otherwise financially sound but just looks to “hand over the keys” when the value of the housing asset has fallen below the outstanding amount of the mortgage. Once enforced, a deficiency judgment can last for up to 20 years.

Statutory Right to Redemption

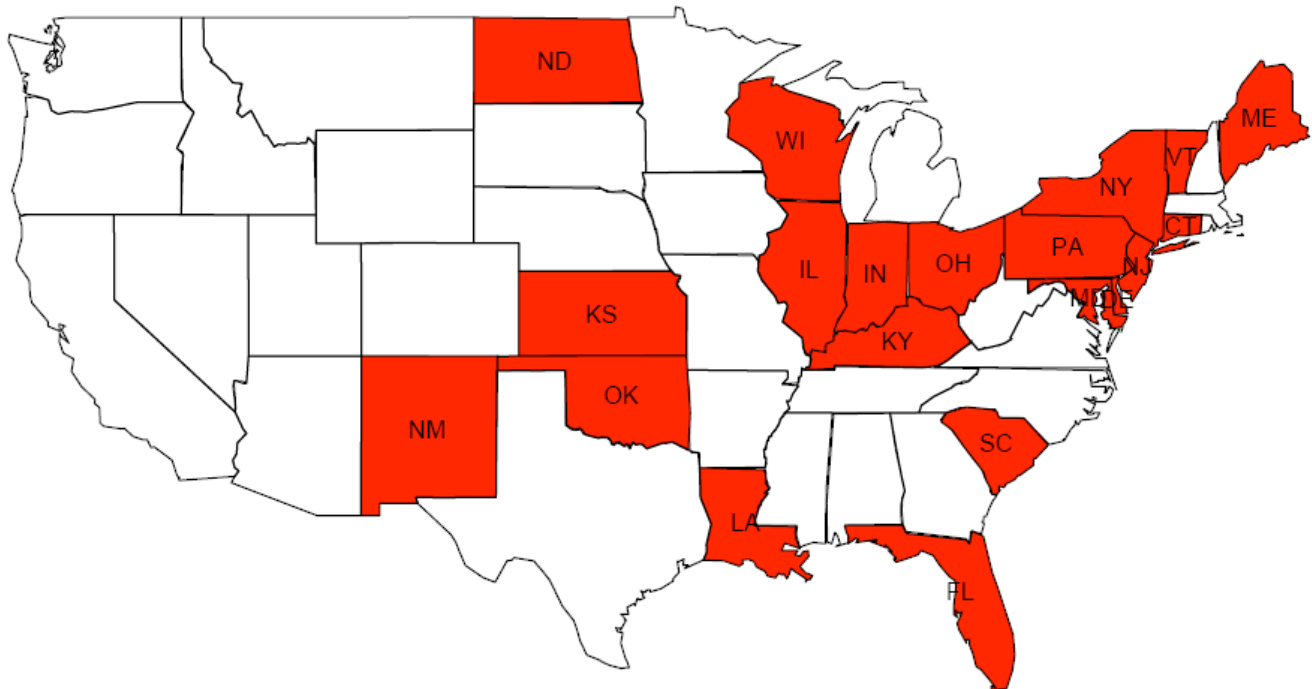
This right provides a homeowner with the ability to buy back a property for a limited period of time (the **redemption** period) after the foreclosure sale (for the foreclosure sale price plus expenses). The homeowner is typically allowed to stay in the house over the redemption period. If they are unable to repurchase the property by the end of the period, the purchaser at the foreclosure sale receives title to and possession of the property.

The statutory right to redemption is not available in all states and various sources list anywhere between 9 to 17 states providing this right (Figure D.3 is based on Pence (2003) and shows 9 states). The confusion seems to arise from the fact that (a) the law keeps evolving and (b) states with a redemption period of less than twelve months are sometimes counted as not having this right.

The motivation behind the right of redemption is to prevent the lender’s agent or other participants in the foreclosure sale from “back-bidding” the property. However, the right appears to also have the side-effect of repelling people from participating in the foreclosure sale since the borrower has the right to stay in the house over the redemption period and also has the right to buy it back.

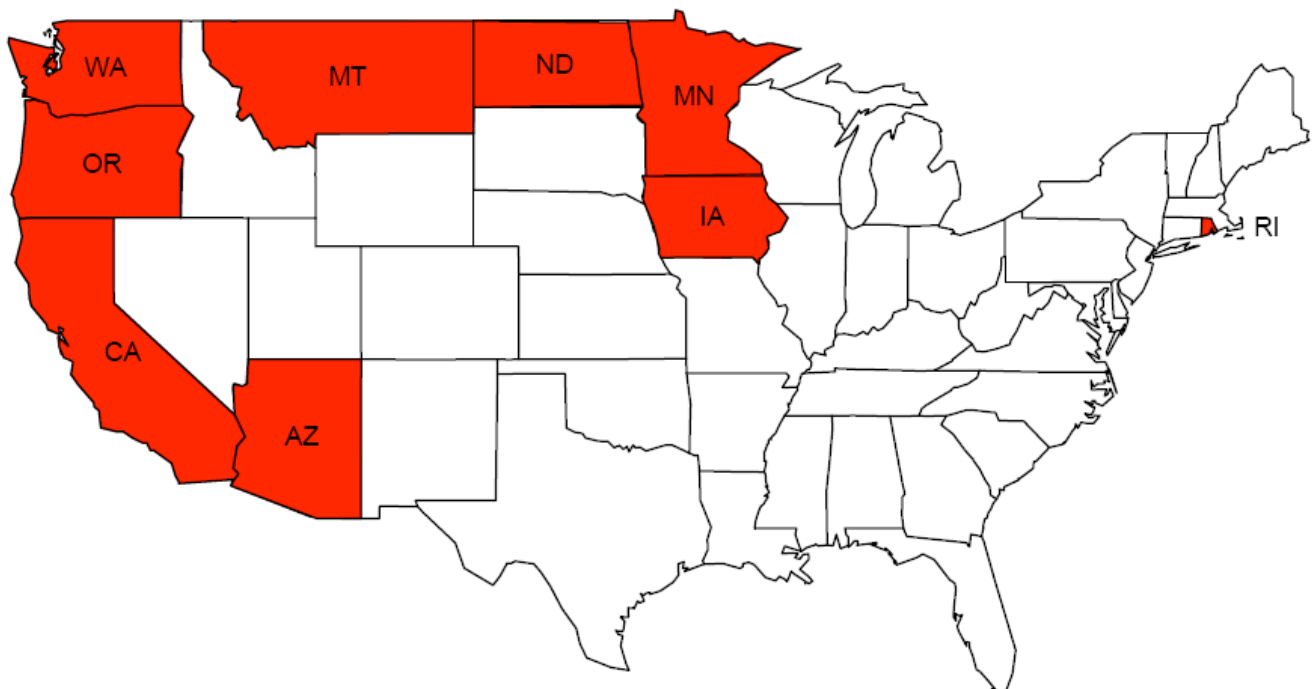
⁴⁰ The empirical research on this topic suggests that in the past lenders have sought deficiency judgments in only 3 to 7 percent of all foreclosure sales. See Nelson and Whitman (2004).

Figure D.1: States Requiring a Judicial Foreclosure Process

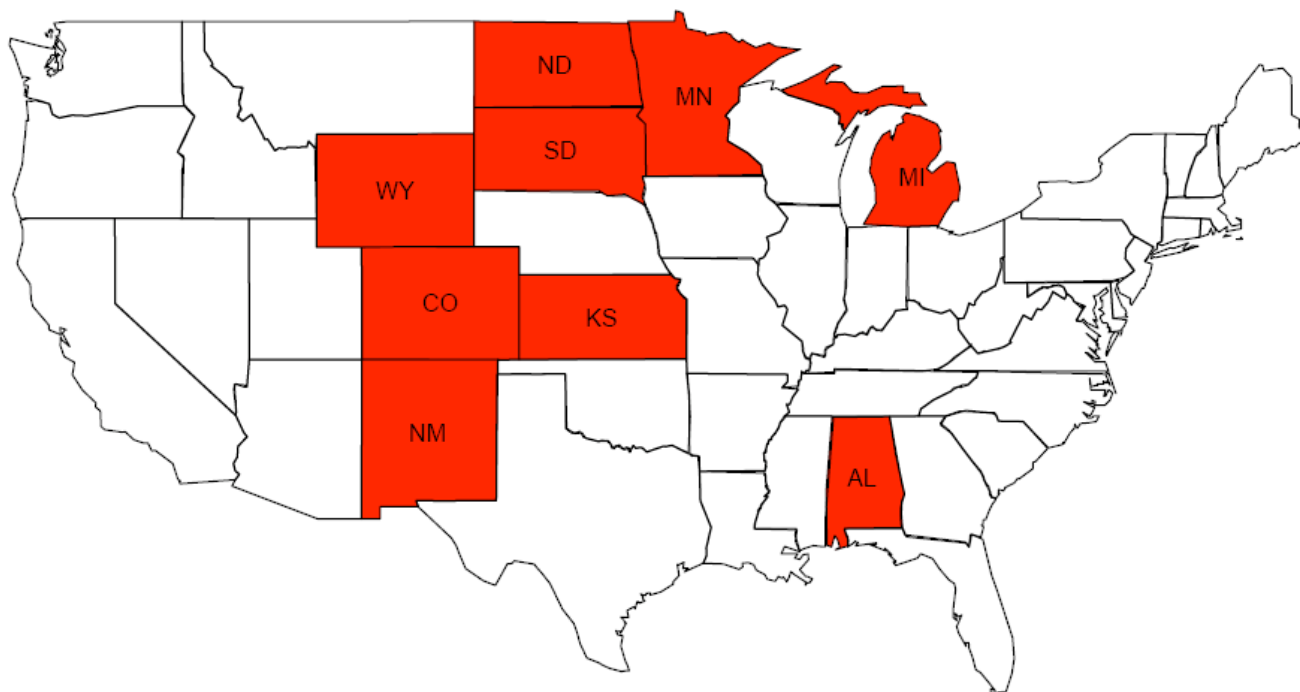


Source: Pence (2003)

Figure D.2: States Prohibiting Deficiency Judgments



Source: Pence (2003)

Figure D.3: States Requiring a Statutory Right of Redemption

Source: Pence (2003)

APPENDIX E. THE ROLE OF JUNIOR LIENS IN MORTGAGE DEFAULT

Introduction

As is widely known now, the boom in home values over 2003-2006 and the resultant weakening in lending standards led to many American homeowners leveraging their homes at much higher levels than before. Since first mortgages with LTVs above 80% require mortgage insurance in the U.S., many homeowners found it more cost effective to liquefy their home equity by getting a second mortgage. This second mortgage was either a closed-end (“HELoan”) or open-end (“HELOC”) second mortgage. The second mortgage industry is often referred to as “home equity lending” and the lenders originating these loans are often different from the mortgagees corresponding to the first loan.⁴¹ Figures E.1 and E.2 show just how prevalent home equity loans have become in the U.S with our estimates in Figure E.1 suggesting that perhaps as many as 4 out of 10 of all mortgaged homes carry a second mortgage.⁴² Figure E.2 clearly shows how the use of these second mortgages became much more pronounced in recent years.⁴³

While the legal framework governing the relationship between the first and second mortgage is typically taken for granted, the nuances of this framework are much more important in the current environment given the preponderance of junior mortgages and the sharp decline in home values that has occurred over the past year. In today’s environment, many 2nd mortgage lenders have already written down most of their position and have little or nothing to lose by waiting for home prices to rise, thus recovering some of the amount they are owed. On the other hand, first mortgage lenders still have significant amounts of exposure and are economically incented to pursue loss-mitigation related measures with distressed borrowers such as short sales or short refis that will maximize the amount of the loan that they recover. Since these sales would in most cases extinguish the value of the call option second mortgage lenders have on home prices, we have a conflict of interest between first and second mortgage holders. We now sketch the legal framework that governs how these conflicts get resolved.

⁴¹ To somewhat confuse matters, not every HELOC is a junior mortgage – we estimate that approximately 20% of all HELOCs are first liens.

⁴² Or a junior lien in general. Some houses may have even have third or fourth mortgages on them.

⁴³ The numbers in Figure E.2 suggest a lower number of second liens than the estimate we arrived at Figure E.1. There are several possible reasons for the discrepancy. One explanation is that the Loan Performance database only records the presence of a second mortgage at the initiation of the first mortgage and does not capture any subsequent increases (or decreases for that matter) in leverage by the borrower. The second explanation could be that seconds are under-reported in securitizations.

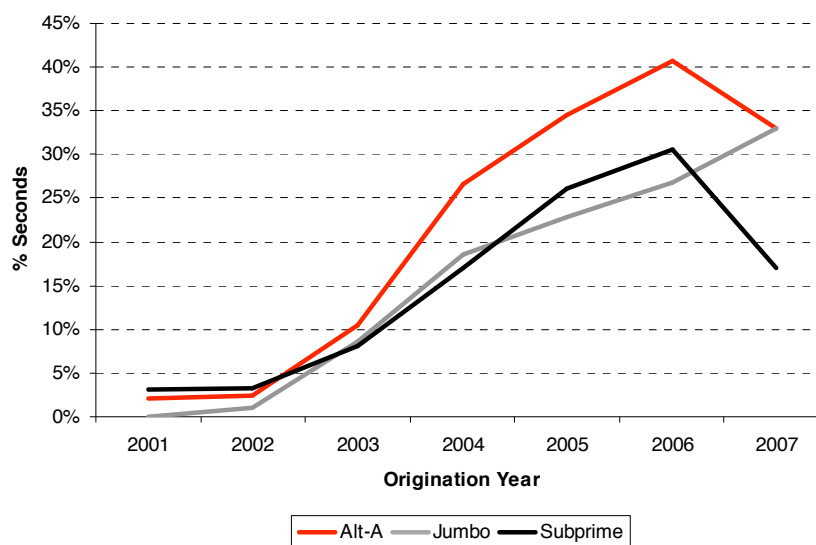
Figure E.1: The Fraction of Mortgaged Single-Family Homes with a Junior Lien

	Volume (\$bb)	Avg. Loan Size (\$K)	Count (#)
HELoans	277	55	5,036,364
1st-lien HELOCs	167	47	3,553,191
2nd-lien HELOCs	668	44	15,181,818
Closed-end Firsts	9,406	200	47,030,000
Total SF Mortgage Debt	10,518		70,801,373

Number of 1st liens	50,583,191
Number of 2nd liens	20,218,182
%Seconds	40%

Source: Banc of America Securities LLC estimates based on data from the Federal Reserve, Inside MBS&ABS, and Company reports. As of December 2007.

Figure E.2: %Seconds in the Non-Agency Market By Sector and Origination Year



Source: Loan Performance, Banc of America Securities LLC.

Lien Priority

To make the discussion concrete, let's consider a representative situation. Suppose a homeowner first takes out a loan for \$150,000 and then follows up by establishing a HELOC for \$50,000. What happens if the homeowner wants to pay off the first lien? Some instances in which something like this might happen is if the mortgage rate on the first lien has fallen while remaining relatively static on the second lien or, in a situation which is quite likely to occur nowadays, if the lender wants to execute a **short refinance**⁴⁴ by writing down all of the HELOC and some of the first mortgage. The key legal concepts that come up in this situation are **subrogation** and **resubordination**.

Resubordination simply means that when the first mortgage is refinanced, the holder of the original second mortgage agrees to resubordinate their loan to second place again. The process for doing so usually requires the junior mortgage lender to sign an agreement agreeing to the

⁴⁴ See Section III on Loss Mitigation for further details.

⁴⁵ For further details, see Grant S. Nelson and Dale A. Whitman, *Adopting Restatement Mortgage Subrogation Principles: Saving Billions of Dollars for Refinancing Homeowners*, Brigham Young University Law Review (2006), p. 305.

terms of the first mortgage refinancing. Is resubordination something that happens automatically in the situation we have discussed above? The answer varies by state and is in general “No.”⁴⁵

To understand how state property law handles resubordination, we need to discuss the legal concept of subrogation. This principle comes into play when the proceeds of a new mortgage are used to pay off a preexisting mortgage. In real estate law, the principle of subrogation states that resubordination is automatic, unless the subrogation would result in “material prejudice” to the existing second. In plain English, subrogation says that whenever the first lien mortgage is refinanced, the second mortgage does not assume the first position and this position is instead taken by the mortgage that replaced the original first mortgage. Exceptions exist where it is clear that the terms of the new first mortgage are increasing the default risk being faced by the second lender. For example, this might happen if the refinanced first mortgage has a higher balance (because of a cash-out refinance) or a higher rate.

If every state were to follow the principle of subrogation then life would be simple but in fact the majority don’t.⁴⁶ There are in practice three approaches taken by the states. The minority automatically resubordinate the second mortgage (in other words they follow the principle of subrogation) unless, as discussed above, the new loan increases the loss exposure of the second. The majority of states stipulate that the previously subordinate lien will not be resubordinated if the refinancing lender has **actual knowledge** (“direct and clear evidence”) that other liens exist on the property. A third set of states takes a hard line that denies subrogation even if the lender only has **constructive knowledge** (“knowledge that could be gathered through ordinary diligence”) of other liens.⁴⁷

With all these legal preliminaries out of the way, the upshot of our discussion is that paying off the first lien requires a negotiation between the first lender and the second lender in most states. In particular, these negotiations are unlikely to reach quick and satisfactory conclusions in today’s distressed housing environment where the interests of first and second lenders are diametrically imposed. This in turn imposes significant hurdles on short sales/short payoff-related loss mitigation activities.

Clearly, the principle of subrogation has important ramifications in default situations since the proceeds from the sale of the house are initially applied towards paying off the first lien with any remaining amounts being used to satisfy the second. Under no circumstances does the first lien holder have the authority to write off second lien debt. Therefore, the requirement in the Frank-Dodd bill regarding extinguishing all junior liens before refinancing can only be met by working out an agreement with the second lien holder.

⁴⁶ This paragraph is based on Nelson and Whitman (2006).

⁴⁷ Unfortunately, we do not have a comprehensive list of which states follow which approach but Texas, New York and California follow the actual knowledge subrogation approach while Florida, Connecticut and Michigan follow the constructive knowledge subrogation approach.

APPENDIX F. PERSONAL BANKRUPTCY

Definitions and Basic Concepts⁴⁸

Bankruptcy is a formal legal procedure for providing a “fresh start” to individuals (and corporations) who are unable to service their debts based on their current level of income. The motivating factor behind the existence of this law is to allow the “honest but unfortunate debtor” a new beginning without being overwhelmed by their existing debt burden. Thus, the bankruptcy process allows debtors to cancel most of their unsecured debts (“Chapter 7” bankruptcy) or construct a repayment plan that pays existing creditors through future income over a three or five year period (“Chapter 13” bankruptcy). One of the most important aspects of how bankruptcy law gives debtors some breathing room is through the **automatic stay** – once an individual files for bankruptcy, this feature of bankruptcy law essentially freezes creditors and collection agencies from pursuing any actions against the debtor or attempting to seize the debtor’s property unless the law or the bankruptcy court allows them to do so. The bankruptcy process ends with the **bankruptcy discharge** (if approved) that releases debtors from liability for specific debts and also prohibits creditors from further collection related activities.

The laws that spell out what types of protection the bankruptcy process provides debtors and creditors are essentially set at the federal level, although the states do have a fair amount of flexibility in terms of defining how much of the debtor’s property is exempt from being sold to satisfy creditors. Bankruptcy cases can be filed in any of 90 federal bankruptcy courts and are mostly handled⁴⁹ by a **bankruptcy trustee**. Basically, the trustee controls the debtor’s property and debts (the **bankruptcy estate**) and has the two-fold responsibility of (1) Seeing that unsecured creditors are paid as much as possible and (2) Making sure the debtor complies with bankruptcy law. Trustees are appointed by the Department of Justice (DOJ) and are in turn supervised by the Office of the U.S. Trustee.

Behind these high-level statements is a fairly intricate set of legal definitions that describes the roles and responsibilities of debtors, creditors, and trustees in the bankruptcy process and tries to ensure fairness for both debtors and creditors. Our goal is to present a reasonably in-depth view of personal bankruptcy because of its importance in determining the credit performance of a range of consumer loans including credit cards, car loans, home mortgages and student loans. For example, in a 1996 study, VISA estimated that between 40% to 45% of all losses on bank credit cards could be attributed to consumer bankruptcies.⁵⁰ Our analysis of bankruptcy begins with reviewing how bankruptcy data is compiled and also discusses some of the pitfalls associated with using this dataset to track the overall rate of personal bankruptcies. Next, we review some of the trends seen in bankruptcy data at the national level and discuss the main drivers of changes in the rate of personal bankruptcies. Finally, we round out our understanding of bankruptcy by focusing on the different bankruptcy options available to individuals along with a quick sketch of what motivates their preference for one choice versus another. We conclude by reviewing and summarizing how different consumer debts are treated in the bankruptcy process.

⁴⁸ For an accessible review of the bankruptcy process and several details about the different bankruptcy chapters see Leonidas Ralph Mecham, *Bankruptcy Basics*, Bankruptcy Judges Division, Administrative Office of the United State Courts (April 2006).

⁴⁹ The bankruptcy court is run by a bankruptcy judge who typically plays a small role in the bankruptcy proceedings. The judge’s rulings are usually limited to eligibility issues and making decisions about whether the debtor receives a discharge.

⁵⁰ See *Consumer Bankruptcy: Causes and Implications*, VISA Consumer Bankruptcy Reports, VISA USA, July 1996.

Personal Bankruptcy Data and its Shortcomings⁵¹

Monitoring bankruptcy trends is challenging because of the lack of accurate and timely data. The core issue is the absence of a central repository of all the information that is available at different stages of the bankruptcy process. As shown in Figure F.1, there are two main bankruptcy reporting agencies: the Executive Office of the U.S. Trustees (EOUST) and the Administrative Office of the U.S. Courts (AOUSC). The EOUST is the executive branch and the AOUSC is the judicial branch of the DOJ. As we discuss below, because of the difference in the statutory mandates of these two agencies, the bankruptcy data reported by both are often difficult to consolidate.

Relevant information on a bankruptcy enters the system through various entry points. A lawyer opens a bankruptcy case in the Electronic Case File system (ECF is the new electronic docketing system for the courts) by providing all the mandatory data relevant to a bankruptcy filing (social security number, total dollar value of property, dollar value of outstanding debts, etc.) related to the debtor. Subsets of data related to each case are collected from each court and transferred to the PACER (Public Access to Court Electronic Records) service center on a nightly basis. Each case is provided with a U.S. Party/Case Index (USPCI), which can be used as a national index by federal, district, and bankruptcy courts. Case opening statistics are extracted and made available through PACER. As the case progresses, the courts' electronic docket system should in theory be updated but the updating practices are not uniform across all courts and, consequently, no aggregated information (like the fraction of case discharges) is compiled on cases in progress. Nevertheless, the progress of each individual case is available on the PACER system.

The U.S. Trustees gather case-related and docketing information from the ECF and transfer it to their older database (ACMBS) with limited fields. Recall that U.S. trustees are responsible for making sure that the bankruptcy system is working as planned and consequently collect a wealth of information that is used to evaluate the performance of bankruptcy trustees and to also calculate the total cost of each case. For example, the U.S. trustee calculates the amount of money received and disbursed to each creditor class. Unfortunately, the information thus collected is decentralized and is not made publicly available. In fact, only a small amount of the information is reported to EOUST, where it is aggregated to compile national statistics.

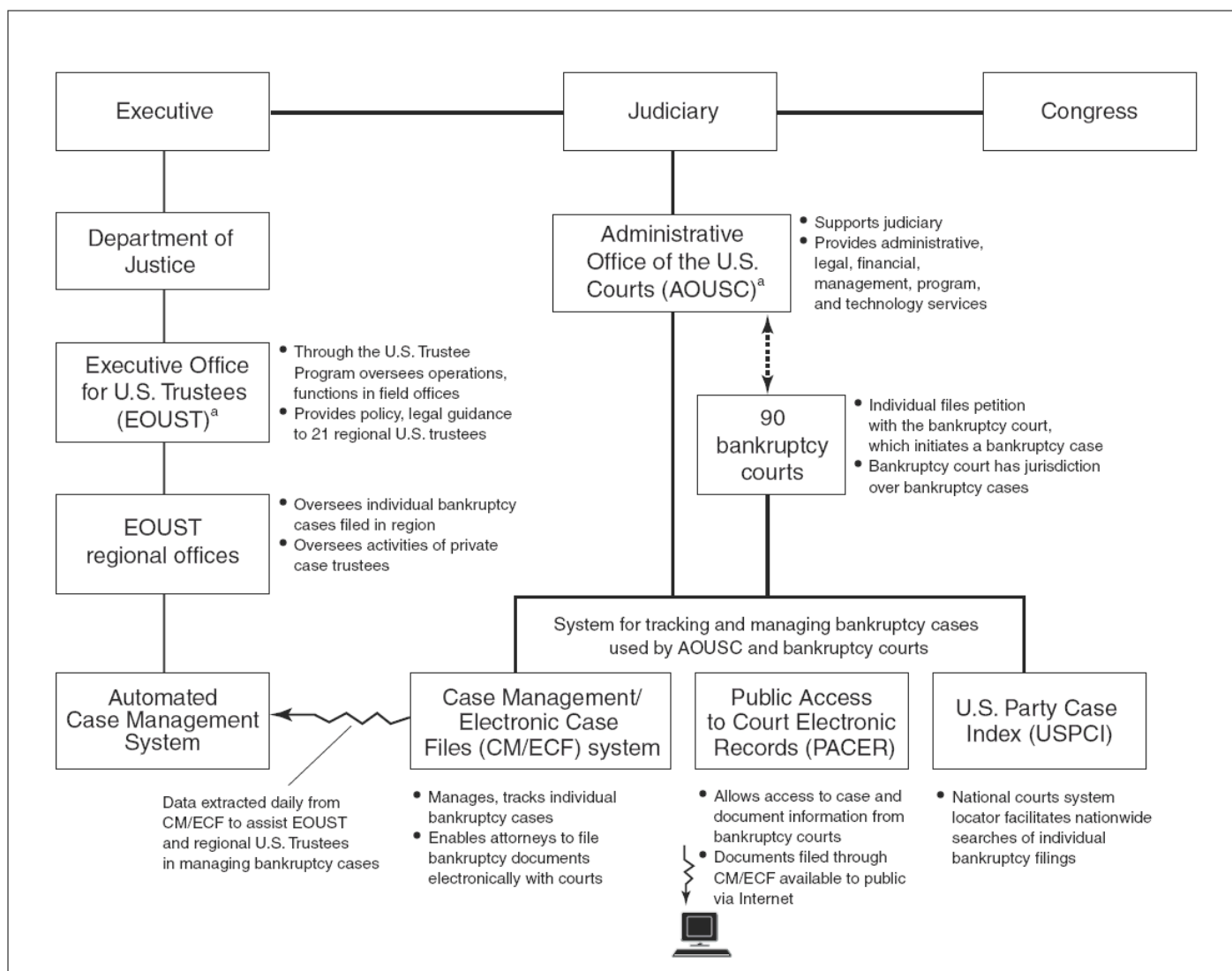
In a nutshell, the problem with bankruptcy data is that case opening information is the only data that is uniformly recorded across all courts and is hence the most commonly used data for many bankruptcy studies. The problem with using this data to assess bankruptcy rates is analogous to the issues we face when we try to project mortgage origination volumes from mortgage application data. First, many mortgage applications may not close because the borrower is rejected by the lender. Second, there is a double-counting issue since many borrowers will file multiple applications and pick the lender who offers the best terms. The analogy with bankruptcy data is fairly precise because the court may reject a filing and also because a single debtor may have multiple filings. Within the context of bankruptcy, multiple filings can result from **serial filings** and **face filings**. Serial filings are repeat Chapter 13 filings by individuals who are unable to make their current repayment plan work. A face filing is when a consumer files for bankruptcy in order to invoke the automatic stay provision of bankruptcy law but has no intention of following through on the filing.

There are still further issues with the data. One concern is the overall reliability of the information that debtors present in their case filings especially with respect to the value of their

⁵¹ Our discussion is partly based on the National Bankruptcy Review Commission Final Report: *Bankruptcy: The Next Twenty Years*, October 1997.

property and existing liabilities. Another shortcoming is that personal bankruptcy data will be contaminated by small business filings since the owners of these businesses can file for a non-business bankruptcy as they often rely on their individual property and credit lines to fund themselves.⁵²

Figure F.1: Overview of the Federal Bankruptcy System



Source: GAO.

^aSix districts in Alabama and North Carolina do not have regional trustees and are not under the jurisdiction of the Executive Office for U. S. Trustees. Instead, these six districts have bankruptcy administrators who are under the jurisdiction of the federal judiciary. The bankruptcy administrators carry out duties similar to those of the regional U.S. Trustees, including the administration of bankruptcy cases, maintaining a panel of private trustees, and monitoring the transactions and conduct of parties in bankruptcy.

Source: GAO

⁵² See Congressional Budget Office, *Personal Bankruptcy: A Literature Review*, September 2000. The paper suggests that small business related bankruptcy filings could account for as many as 6%- 20% of all personal bankruptcy filings.

The Macrostructure of Personal Bankruptcy

Despite all the drawbacks associated with using the case openings (filings) data identified above, it is the only broad-based data set available over an extended time period and is thus commonly used to assess trends in personal bankruptcy rates. At a broad conceptual level, changes in these trends (bankruptcy filings measured on a per-capita basis) can be attributed to the following three factors:

1. **State and Federal Bankruptcy Law:** Changes in law that make it easier or more difficult to file for bankruptcy.
2. **Economic factors:** Shocks to income through job loss, increases in household leverage, expense shocks (utilities, food, medical expenses), an insufficient buffer of personal savings, etc.
3. **“Social Attitudes”:** Changes in the stigma associated with bankruptcy, consumers looking at filing for bankruptcy in a more strategic way because of legal advice.

Obviously, it's not easy to disentangle the standalone contribution of these factors to bankruptcy rates and, in fact, most of the academic research has sent conflicting signals on the relative importance of different variables that have been proposed as causative factors.⁵³ Still, we will provide a broad sketch of how different factors affect the bankruptcy rate, although it is worth re-emphasizing that the bankruptcy rate appears to have different sensitivities to these factors over different time periods.

To begin with, take a look at Figure F.2 which shows that the overall personal bankruptcy rate was steadily trending upward between 1985 and 2004, before rising and falling sharply in 2005. While changes in bankruptcy law are relatively infrequent, the 2005 bankruptcy data do conclusively demonstrate the importance of these laws in determining the filing rate since the upward and downward spikes in the graph respectively correspond to the period prior to and after the introduction of the Bankruptcy Abuse Prevention and Consumer Protection Act of 2005 (BAPCA) on October 17th, 2005. Some of the background behind the introduction of this law is that the most common type of bankruptcy filings (Chapter 7 filings) averaged 70% of total filings and the common perception was that many of these Chapter 7 filers were cancelling debts that they were actually able to repay.⁵⁴ Among other reasons, BAPCA was introduced to make it more difficult to file for a Chapter 7. In anticipation of the enactment of the law, Chapter 7 filings and hence total filings shot up and then dropped equally sharply immediately after the law's introduction. Some of the most important features of BAPCA versus prior bankruptcy law are:

- It is now harder for debtors to qualify for a Chapter 7 bankruptcy, thus effectively forcing more people to file for a Chapter 13 bankruptcy.
- BAPCA increases the required waiting period between two Chapter 7 filings to eight years.
- BAPCA requires debtors to undergo counseling in credit and budget management through a government-approved non-profit.

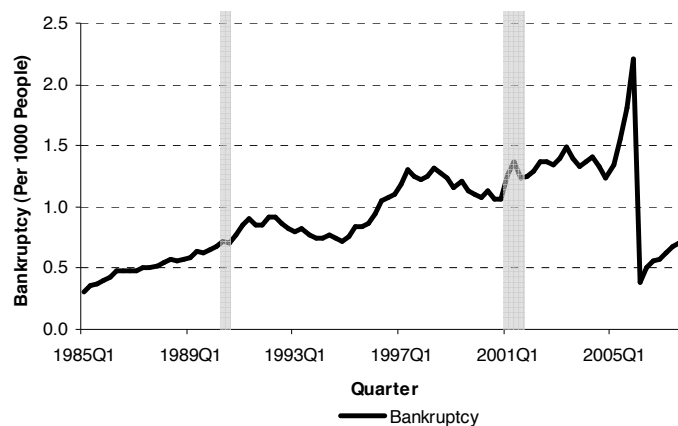
⁵³ A summary of this research can be found in the CBO paper cited in the previous footnote. Also see Thomas A. Garrett, *The Rise in Personal Bankruptcy*, Thomas A. Garrett, Federal Reserve Bank of St. Louis, October 2006.

⁵⁴ We will discuss the different types of bankruptcy options in detail later on.

- Bankruptcy attorneys are now required to certify to the accuracy of debtor filings.
- Bankruptcy filings are now subject to a greater number of audits by U.S. Trustees.

Another meaningful change in bankruptcy law (over the period graphed in the figure) occurred through the Bankruptcy Reform Act of 1994 which increased the amount of assets that debtors could retain in a Chapter 7 filing. This change decreased the costs associated with filing for bankruptcy and partially accounts for the increases in the bankruptcy rate in 1994-1995 during a period of economic strength. We will also discuss some of the other factors that contributed to this increase over the next few paragraphs.

Figure F.2: Trends in Personal Bankruptcy Filings: 1985 - Present

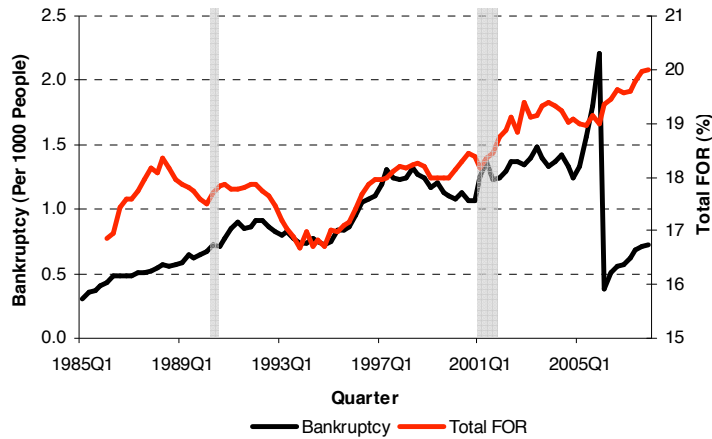


Note: Recessions are denoted with grey bars

Source: American Bankruptcy Institute

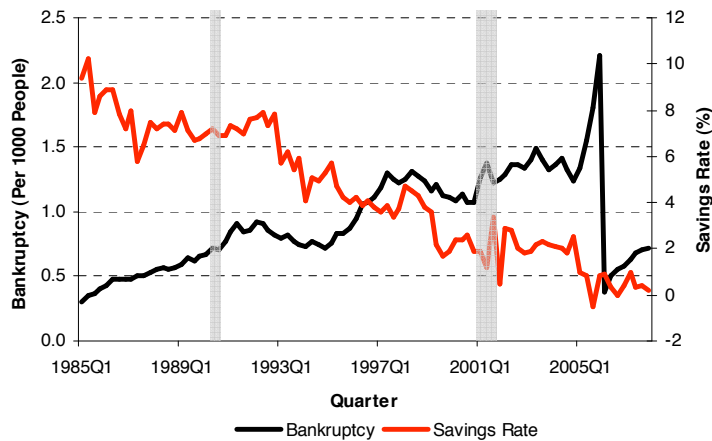
Moving from the legal to the economic drivers of personal bankruptcy, note that many of these economic factors will capture a measure of household default risk. For instance, some household default risk factors include the total amount of indebtedness, the average cost of servicing this debt, shocks to income, depletion of personal savings, or increases in living expenses. A debt-service ratio calculation bundles many of these factors into one calculation. For example, the Federal Reserve's Financial Obligations Ratio (FOR) is the ratio of the sum of payments on outstanding mortgage and consumer debt, automobile lease payments, rental payments, homeowner's insurance and property tax payments to disposable personal income. Figure F.3 shows that since the 1990s, the trend in bankruptcy filings closely follows the overall evolution of FOR and starts to diverge beginning in 2000 when high levels of home equity buildup allowed homeowners to consolidate credit card debt into their mortgage balances and to also use home equity gains to pay for expenses. This explains why bankruptcy rates remained relatively stable over the 2000-2005 period even though FOR was trending upwards. Figure F.4 also shows how the personal savings rate steadily decreased over the same period, although the impact of this trend was somewhat mitigated by an increase in the value of household assets (not shown in the figure) over the same time frame.

Figure F.3: FOR Captures Several Factors that Affect the Default Risk of a Household (Lagged by 4 Quarters)



Note: Recessions are denoted with grey bars
Source: Federal Reserve

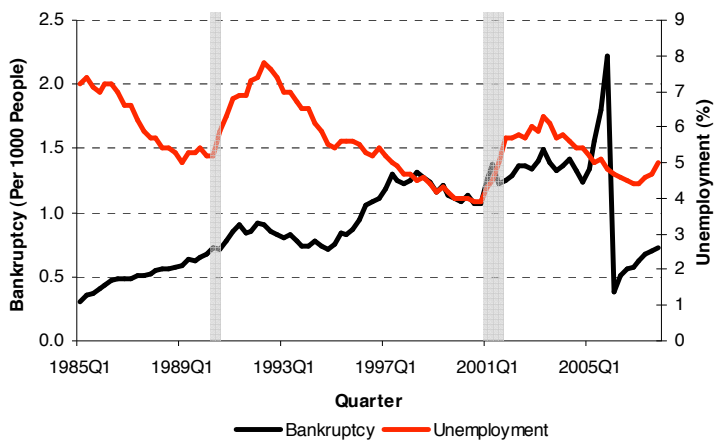
Figure F.4: Personal Bankruptcy Driver: The Savings Rate



Note: Recessions are denoted with grey bars
Source: Bloomberg

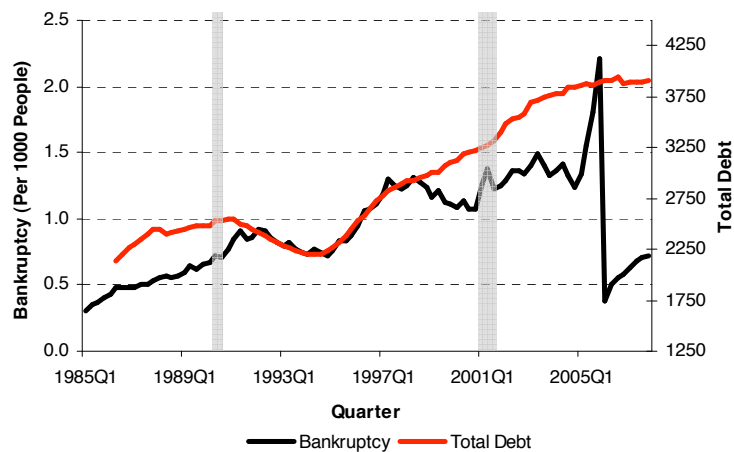
Ascertaining the impact of other variables on the bankruptcy rate is more challenging as the importance of these factors appears to vary over time. For instance, we juxtapose the bankruptcy data from Figure F.3 with several variables that measure household default risk including unemployment rates (Figure F.5), total indebtedness (Figure F.6), Willingness to Lend (Figure F.7), Revolving Debt (Figure F.8) and Non-Revolving Debt (Figure F.9). As we can clearly see from examining these figures, while there are periods where bankruptcy trends and each of these risk measures move side by side, there are also significant stretches where the two series decouple, thus speaking to our earlier point of the challenge in quantitatively assessing the sensitivity of the bankruptcy rate to changes in these risk factors. Still, one other broad generalization we can make is that personal bankruptcy trends tend to be strongly influenced by the “business cycle effect,” particularly when the cycle is consumer-driven. Thus, we typically see bankruptcies increase in recessions as lending tightens and unemployment increases and fall as the economic environment improves.

Figure F.5: Personal Bankruptcy Driver: Unemployment Rate



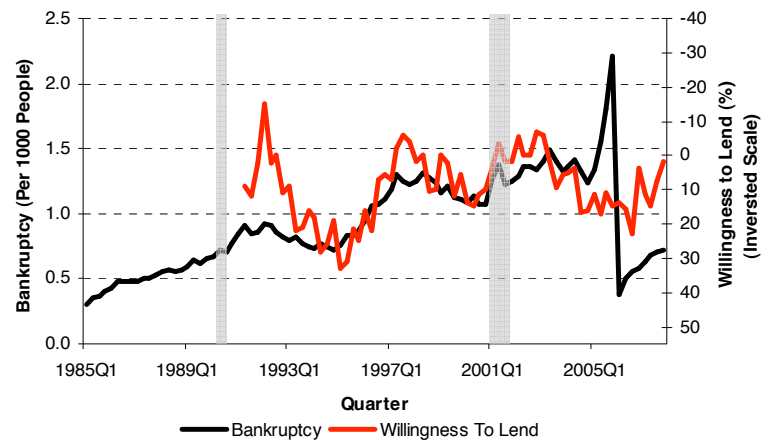
Note: Recessions are denoted with grey bars
Source: Bloomberg

Figure F.6: Personal Bankruptcy Driver: Per Capita Total Debt (Lagged by 5 Quarters)



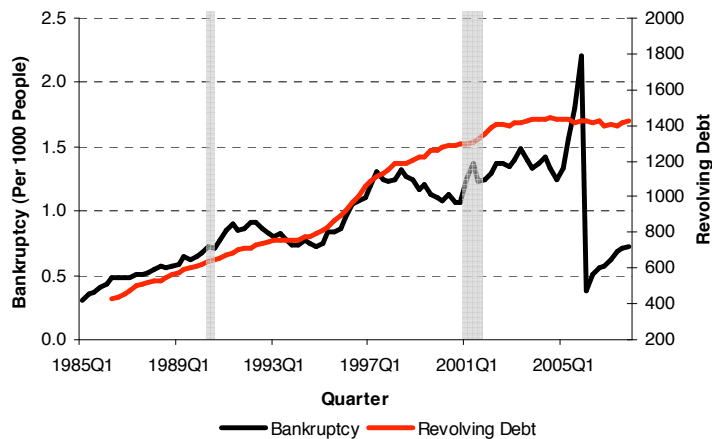
Note: Recessions are denoted with grey bars
Source: Bloomberg

Figure F.7: Personal Bankruptcy Driver: Willingness to Lend Consumer Installment Loans (Lagged by 5 Quarters)



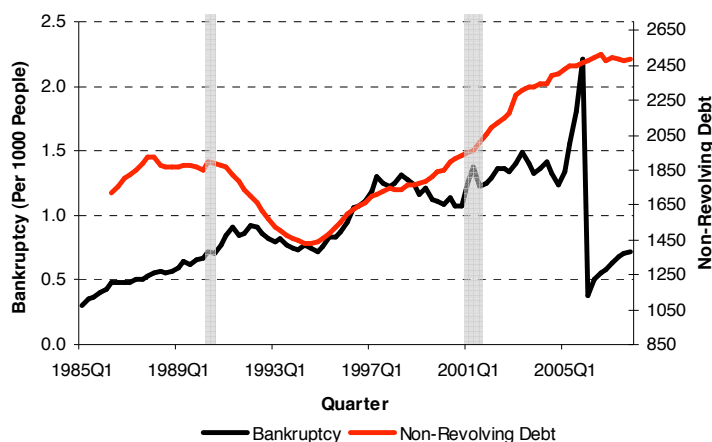
Note: Recessions are denoted with grey bars
Source: Federal Reserve

Figure F.8: Personal Bankruptcy Driver: Per Capita Revolving Debt (Lagged by 5 Quarters)



Note: Recessions are denoted with grey bars
Source: Bloomberg

Figure F.9: Personal Bankruptcy Driver: Per Capita Non-Revolving Debt (Lagged by 5 Quarters)



Note: Recessions are denoted with grey bars

Source: Bloomberg

However, even here (the “business cycle effect”) there are counterexamples with the period from 1994 to 1997 being particularly interesting because of a persistent increase in bankruptcies over a period of rapidly improving economic standards. As mentioned previously, one of the contributors to this trend was a relaxation of bankruptcy laws. Also cited as a cause is an expansion in consumer lending to borrowers with weaker credit (notice the increase in consumer debt over the relevant period in Figures F.8 and F.9). Finally, several researchers have attributed this anomaly to one of the high-level cluster of factors we cited as drivers of bankruptcy – changes in the social attitudes of consumers. Some studies have argued that coincident with the increase in the number of bankruptcies, individual awareness about the bankruptcy process improved because of greater advertising by lawyers, coverage by the media, and knowledge sharing among family and friends. This “awareness” thus potentially reduced the stigma associated with bankruptcy and made consumers more confident about increasing their leverage, thereby also effectively increasing their chances of bankruptcy. Changes in the stigma associated with bankruptcy are still controversial in part because of the difficulty of defining proxy variables that measure this effect precisely.⁵⁵

Of course, there are several other variables that have been also been attributed as predictive of bankruptcy rates but they generally fall into the three categories we identified above (macro, legal, social). What makes predicting bankruptcy rates especially challenging is trying to quantitatively assess the contribution of these variables to the overall rate.

The Microstructure of Personal Bankruptcy⁵⁶

At this point we have a working understanding of the different factors contributing to individual bankruptcy and now narrow our focus to trying to understand the legal options available for an individual who is considering filing for bankruptcy and to also discussing how this individual might evaluate the relative attractiveness of their different choices. We also provide some details

⁵⁵ See the CBO study for further details.

⁵⁶ For further details, see Stephen Elias, *The New Bankruptcy – Will it work for you?*, Nolo, June 2007 (2nd Edition).

about how consumer assets are treated in the different types of bankruptcy since this issue is of special interest to investors who purchase consumer debt through participating in the ABS and MBS markets.

In the U.S., the bankruptcy filer has to pick a “Chapter” when filing for bankruptcy. At a very high level, the different bankruptcy Chapters can be described as follows:

1. **Chapter 7.** Some of the debtor’s property (“non-exempt” assets) is sold in order to satisfy creditors. Most of the remaining debt is cancelled. As a result, most unsecured pre-bankruptcy debt is discharged in a Chapter 7 bankruptcy. This is the most common type of bankruptcy and in 2007 over 60% of all bankruptcy filings were Chapter 7s.
2. **Chapter 11.** This is a traditional corporate bankruptcy that is geared towards continuing to operate a business while paying creditors through a court-approved debt restructuring plan. This Chapter can also be used by individuals with large debts but is rarely so utilized because it is expensive and complex.
3. **Chapter 13.** A bankruptcy plan for individuals that is modeled after Chapter 11 – in other words, a 3 or 5 year repayment plan is crafted to payoff existing debts out of the debtor’s future income. No liquidation of assets occurs.

Other less commonly used chapters include:

4. **Chapter 9.** Debt reorganization for municipalities (cities, towns, villages, taxing districts etc.) that is similar to Chapter 11.
5. **Chapter 12.** Debt reorganization for farmers and fishermen through a process that is very similar to Chapter 13.
6. **Chapter 15.** This applies to cross-border insolvency cases and is applicable where the debtor or its property is subject to the laws of the United States and one or more foreign countries.

Going through the list of Chapters above, we can see that conceptually there are two kinds of bankruptcies – **liquidations** (Chapter 7) or **reorganizations** (Chapters 9, 11, 12, 13). The overwhelming majority of all personal bankruptcy cases that occur are Chapter 7s or 13s. Shortly, we will describe these two different chapters and how individual debtors choose between them in detail but, before doing so, let’s foreground some of the important issues and questions since it is easy to get overwhelmed by the legal minutiae of these two different types of bankruptcies. Some of the important questions are:

- Who is eligible to file for bankruptcy?
- Which debts can be discharged in a bankruptcy and which cannot?
- To what extent can a borrower’s assets (“property”) be used to pay off creditors?
- How are repayment plans set up under Chapter 13?
- How does a consumer choose between Chapter 7 and Chapter 13?
- How are the various different types of consumer debts (mortgages, student loans, credit cards, auto loans) treated in the two different types of bankruptcies?

Chapter 7 Bankruptcies

Chapter 7 offers a quick (3-6 months on average), straightforward, and effective way to discharge most unsecured debts and in many cases debtors can retain most of their secured assets. However, starting with the introduction of BAPCA, it's a little more challenging to qualify for a Chapter 7 than before. One of the crucial aspects of the new bankruptcy law is the **means test** – this is a check to see if a debtor is eligible to file for a Chapter 7 or whether they should be trying to pay their debts through a Chapter 13. If a debtor fails the means test, they have to file under Chapter 13.⁵⁷

In general, a debtor doesn't have to take the means test if they meet the following requirements:⁵⁸

- Their average income (as defined below) for the past 6 months is less than the median income for the state of residence; or,
- More than 50% of the debt is non-consumer related; or,
- The debtor qualifies for the disabled veteran exception.

Basically, the means test tries to measure how much money a debtor has left over to satisfy their debts after subtracting living expenses from their most recent income levels. Here are some of the factors that go into the calculation:

- (+) Current monthly income (includes taxable and nontaxable income, alimony/child support but not Social Security benefits);
- (-) Allowable monthly expenses determined by the IRS in 3 basic categories: (1) food, clothing, and other items; (2) housing and utilities; and (3) transportation;⁵⁹
- (-) Spending on dependent members of the debtor's household;
- (-) Spending for health insurance, disability insurance, HSAs;
- (-) Primary or secondary education of a dependent child;
- (-) Average monthly payments on secured debts (home mortgages, car loans);
- (-) Average monthly payments on **priority claims** (child support, student loans, alimony);
- (-) Administrative costs.

The result of this calculation is called “disposable income.” A debtor does not qualify for Chapter 7 if their projected disposable income is greater than \$182/month, or if their disposable income is at least \$110/month and they can pay up to 25% or more of their unsecured non-priority debts over a 5 year period. An inability to qualify for a Chapter 7 means of course that they would have to file under Chapter 13.

If a debtor does qualify for a Chapter 7, the basic idea is to start by dividing their assets into two

⁵⁷ A sense for how much BAPCA has changed the landscape of personal bankruptcy is given by the fact that the share of all bankruptcies that corresponded to Chapter 7 filings consistently hovered around 70% prior to 2005 but reached as much as 80%-85% near the introduction of BAPCA. Since then, the percentage of Chapter 7s has hovered around 60%. Of course, as Figure F.2 shows, the overall level of bankruptcies has sharply fallen and then bounced back. There is still a vigorous and unresolved debate as to whether filing rates will return to their pre-BAPCA levels (after “controlling” for the economic environment).

⁵⁸ It turns out that a significant number of bankruptcy filers do not have to take the means test. According to a survey of 11,000 bankruptcies filed before BAPCA took effect, only 15% of all Ch.7 filers would have had to take the means test. See Elias (2007) for more details.

⁵⁹ These vary according to the size of the household, gross monthly income and other variables. See Elias (2007) for further details.

buckets: “exempt” and “non-exempt.” Then, the bankruptcy trustee sells non-exempt assets to satisfy unsecured creditors.⁶⁰ All remaining pre-bankruptcy unsecured debt is discharged and creditors do not have any claim on future earnings.

This leaves us with two questions: What assets are exempt versus non-exempt and what happens to secured debts? Let’s discuss exemptions first since they are a prerequisite to understanding the treatment of secured debt. The area of exemptions is one place where bankruptcy law becomes quite complicated. Exemptions exist at the state and federal level but states are allowed to “opt-out.” Currently, 35 states make you choose the state-level exemption whereas the debtor has flexibility to choose between the federal and state exemption in the remaining states. Needless to say, the exemptions available from state to state are far from homogeneous. In general, property that is considered necessary to support a basic standard of living (furniture, clothes) is exempt from being sold to satisfy creditors regardless of value. Other items including homes and automobiles are exempt up to a capped value (we will elaborate on this point shortly).⁶¹

Non-exempt items (i.e., property that can be sold to pay unsecured creditors) includes investments (cash, deposits, stocks, bonds), expensive musical instruments unrelated to the debtor’s job or business, hobby collections etc.

To understand what happens to secured debt, we need to understand how the property that collateralizes this debt is treated. In general, property that is secured by a loan (a home, a car) is treated differently than property owned outright:

- If the amount of equity (the value of the property less amount of loan) is less than or equal to the exemption amount, then the debtor can keep the property as long as they can keep making payments on the loan (this is called **reaffirmation**).
- If the amount of equity is greater than the exemption amount, then the property is sold to pay off creditors. In detail, the proceeds from the sale of the property are first used to payoff the existing loan, give the borrower the exemption amount due to them, and then pay unsecured creditors with the remaining amount. However, practically speaking, there has to be some “margin” in the deal for the trustee to seize the property since there will be some incremental costs associated with storage of the item and the actual sale transaction itself.
 - **Example.** The debtor owns a car that is worth \$7,000 and that collateralizes an auto loan of \$3,000. Assume that the exemption amount for cars in the state in question is \$2,000. In this case, the trustee would seize the car and sell it for \$7,000. Out of these proceeds, the auto loan lender would receive \$3,000, the debtor would receive \$2,000, and the remainder (\$2,000) would go to unsecured creditors after accounting for the costs of sale and the trustee’s fees.
- Thus, in the case of secured debt, if a debtor is able to hold onto the underlying property because the amount of equity is less than or equal to the exemption amount, the loan the property is financed with is not discharged in the bankruptcy (as long as the borrower is current on their payments). However, if the borrower falls behind on their payments, lenders can seize the property and sell it.

How does the property valuation process work? This is one area that has changed since the introduction of BAPCA. All assets must now be valued at **replacement cost** (as opposed to

⁶⁰ In practice, almost all Chapter 7s correspond to “no asset” cases – the debtor has no non-exempt property.

⁶¹ Detailed information on state-level exemptions can be found on the website www.bankruptcyinformation.com.

resale value previously) – what it would cost to buy something equivalent. Note that this valuation of the borrower’s property also turns out to be relevant for Chapter 13 even though there’s no sale of assets here. This is because one of the rules associated with Chapter 13 is that the debtor has to pay their creditors at least as much as the creditors would’ve got through the sale of non-exempt assets in a Chapter 7.

So far we have reviewed who qualifies for a Chapter 7, which of the debtor’s assets can be liquidated for the benefit of creditors, and what happens to secured and unsecured debt. We wrap up our review by going into a little more detail about a critical aspect of a Chapter 7 – what types of debts get discharged and which survive. Typically, most unsecured debt gets charged off including credit cards (except in cases of fraud or luxury purchases made right before filing for bankruptcy), medical bills, and lawsuit judgments. Debts that cannot be charged off include domestic support obligations (child support, alimony); debts owed to a spouse, former spouse, child incurred in the course of a divorce or separation; fines, penalties or restitutions from a federal, state or local government; certain tax debts; debt that wasn’t discharged in a previous bankruptcy; student loans; and regular income taxes. Also, secured debt cannot be discharged.

Getting a bankruptcy discharge is not a preordained outcome for a Chapter 7 filer. Although it is pretty rare, an objection to the discharge could be filed by the trustee or a creditor. Typically, this would happen if it were believed that the debtor had not been upfront in terms of declaring all their assets, or that they were hiding their income, or generally engaging in other types of fraudulent activity. However, once the debtor has had their debts discharged in a Chapter 7, they cannot re-file for the same Chapter for another 8 years.

Chapter 13 Bankruptcies

Chapter 13 is a bankruptcy plan for debtors who earn a regular income and who therefore could potentially payoff their creditors somewhere down the road. In exchange, the debtor does not have to give up their assets (including non-exempt assets) since creditors are being paid through future income rather than asset sales.

The key part of a Chapter 13 is the repayment plan. The repayment plan lasts either 3 years or 5 years, depending upon the debtor’s income level. If the debtor’s gross average income over the past 6 months is less than the state’s median income, the repayment plan will last three years. If not, the repayment plan lasts 5 years. Other important aspects of a Chapter 13 are:

- Even if the debtor doesn’t have to repay their debts in full, they still have to pay at least as much as creditors would have received in a Chapter 7 liquidation of non-exempt assets (recall that this is why property valuations are important even for Chapter 13s).
- The repayment plan has to be approved by the bankruptcy trustee. Specifically, the court has to judge that the plan has been filed in “good faith” – in other words, the borrower is genuinely unable to discharge their debts from current income. Once the plan is approved by the bankruptcy court judge, creditors cannot block it.
- Typically, repayment plans involves paying 100% of **priority debt** such as alimony, maintenance, support, income taxes etc and some percentage of the unsecured debts. This percentage depends upon the debtor’s disposable income and the value of their non-exempt property.

The debts that survive a Chapter 13 are mostly the same as the ones that survive a Chapter 7. In

particular, student loans cannot be written off in a Chapter 13.

Getting through a Chapter 13 repayment plan is not a straightforward process and the success rate of these plans is only around 33% (see Figure E.10). Once a Chapter 13 discharge occurs, debtors have to wait four years before receiving a discharge in a Chapter 7, or two years before another Chapter 13.

Figure E.10: Performance of the Chapter 13 Bankruptcy System

	Study #1	Study #2
Repayment plan confirmation rate	82%	77%
Discharge Rate	33%	33%
Recovery rate of secured debt	27%	30%
Recovery rate of unsecured debt	16%	20%
Subsequent refiling rate	30%	33%

Source: Adapted from “What Do We Know About Chapter 13 Personal Bankruptcy Filings,” Wenli Li, Business Review, Philadelphia Federal Reserve (Q4 2007).

Choosing between a Chapter 7 and a Chapter 13

As we mentioned in our discussion of the means test, there’s no notion of a choice between Chapters if you fail the means test – you have to file for Chapter 13 in this situation (or decide not to file for bankruptcy at all). Therefore, let’s consider the cases where a borrower could potentially choose between the two Chapters. If the debtor desires a fast bankruptcy that discharges most debts (except the exceptions we have documented above) then they should pick Chapter 7. If the debtor is current on their loan payments it is actually fairly difficult to lose property in this Chapter because the prevailing exemption laws typically allow debtors to keep most necessities. Still, BAPCA has raised the bar for Chapter 7 by charging higher attorney fees, requiring more documents and counseling, and also imposes tougher state-residency requirements so that debtors are less able to exploit the fact that certain states offer more attractive exemptions than others.

So, why would a debtor choose a Chapter 13? Well, a debtor should pick this Chapter if they are delinquent on their mortgage or car loan and want to make up the missed payments over time and get the loan back to being current, and thus retain the underlying property.⁶² This would not be possible under a Chapter 7 for the following reasons. If the delinquent debtor filed for a Chapter 7 and had more equity than the exemption amount, the trustee would seize the property and sell it to pay creditors. On the other hand, if the amount of equity were less than the exemption amount, the court would typically agree when the (mortgage or auto loan) lender filed a motion to lift the automatic stay associated with a bankruptcy filing. The lender would then complete the foreclosure process and seize the property.

Bankruptcy and the Discharge of Consumer Debts

We summarize the treatment of different types of consumer debts in bankruptcy by reviewing several aspects of our discussion above. The treatment of credit card debt and student loans is

⁶² Another reason for picking a Chapter 13 is that a somewhat broader collection of debts can be discharged. See *Bankruptcy Basics* for more on this issue and other factors influencing Chapter choice.

relatively straightforward in personal bankruptcies. Credit card debt will get charged off. Student loans will survive bankruptcy unless the debtor can demonstrate “undue hardship.” What this means is that the debtor has to show the court that they have little money currently, they expect their poverty to continue, and that in the past the debtor has made a good faith attempt to payoff the loan. Still, courts will rarely discharge student loans.

The principles for home mortgages and car loans are very similar since they both correspond to debt secured by property. We will now discuss some of the details of how homes and mortgages are treated in the bankruptcy process. In general, bankruptcy and housing intersect in two areas:⁶³

1. The ability of a debtor to protect their non-exempt home equity through filing a Chapter 13
2. The ability of the a debtor to stop the foreclosure process by filing for bankruptcy and thus invoking the automatic stay

On the first issue, several states have exemptions available for the amount of home equity (value of the house less the value of the mortgage) in the debtor’s principal residence (the **homestead exemption**). These homestead exemptions can vary wildly from state to state with the exemption amount ranging from a few thousand dollars to being unlimited. The exemptions are only valid for owner-occupied properties and don’t protect investor properties, vacation homes, and second homes. There are also a slew of complicated residency rules with the simplest one being that if you have been in residence in the current home for at least 40 months before filing bankruptcy, then you can use the state’s current homestead exemption. Some indication of the variety of the states exemptions is given by the different categories that they can be organized into:

- States allowing unlimited homestead exemptions;
- Homestead exemptions based on lot size only;
- Homestead exemptions based on lot size and equity;
- Homestead exemptions based on equity only; and,
- States with no homestead exemption.

If a debtor is current on their mortgage, they can keep their home when filing for a Chapter 7 if the amount of equity they have is less than the exemption amount. If a homeowner is delinquent on their mortgage when they file for a Chapter 7, they run the risk of losing their house as the bankruptcy court will probably agree to lift automatic stay in this case. If, however, the homeowner can stave off foreclosure over the period of the Chapter 7 bankruptcy (3-6 months), they may have an easier time servicing their mortgage because their overall debt burden (through the cancellation of credit cards for example) has been reduced.

Delinquent homeowners who wish to retain their house will typically file for a Chapter 13 since the amount they are behind on their mortgage payments can be built into the repayment plan. The question arises, why use a Chapter 13 repayment plan and damage your credit instead of a lender

⁶³ One other area where bankruptcy and housing meet is the area of **cramdowns**. Debtors in a Chapter 13 are allowed to modify mortgages on non-owner-occupied properties (vacation homes, investor properties). This modification can involve changing the interest rate or term of the mortgage or even write down its principal to the value of the underlying collateral (the borrower’s house). This writedown is referred to as a **strip-down** or a **cramdown**. Strip-downs are not permitted on mortgages backed by primary homes since a 1993 Supreme Court decision. For further details, see Adam J. Levitin and Joshua Goodman, *The Effect of Bankruptcy Strip-Down on Mortgage Rates*, Georgetown University Law Center, Business, Economics and Regulatory Policy Working Paper Series, January 28, 2008.

repayment plan? A debtor would prefer a Chapter 13 if, in addition to their mortgage debt, they were also having trouble servicing other debts and want them discharged through Chapter 13. Other possibilities are that the Chapter 13 repayment plan is less onerous than the one suggested by the lender or if there is disagreement with the lender on some key issue. Thus, Chapter 13 is typically used by homeowners who are facing foreclosure and want to “freeze” the foreclosure process. Homeowners can halt the foreclosure process by filing for bankruptcy even after foreclosure starts (with some restrictions).

The final point to keep in mind is that there is no concept of exemption in a Chapter 13, so the debtor can keep the home no matter how much equity they have. However, the constraint is that they owe unsecured creditors at least as much as they could get in a Chapter 7. This means that if they do have a substantial equity above the exemption amount in their home and they desire to retain the home, they will have to come up with funds from alternative sources in a sum equal to the difference between the equity and the exemption amount.

APPENDIX G. THE TAXATION OF MORTGAGE DEFAULTS⁶⁴

As previously detailed in different sections of the primer, some of the costs faced by homeowners upon defaulting include:

- Loss of the house;
- Legal exposure to the lender who can sue for the difference between the amount of the unpaid debt and the foreclosure sale price plus foreclosure expenses (in states that permit deficiency judgments);
- A damaged credit record; and,
- Potential tax liability.

Why do defaulting mortgagors face tax liabilities? The general principle at work here is that the IRS considers forgiven debt to be income. Thus, a lender forgiving the amount of the shortfall between the outstanding debt and the foreclosure sale price (plus expenses) triggers a tax liability for the defaulting mortgagor. The lender is supposed to report the amount of the debt forgiveness on Form 1099-C to the IRS. Debt forgiveness in mortgage lending is actually fairly common because deficiency judgments are rarely pursued against borrowers and in fact are severely circumscribed in nine States.⁶⁵ Of course, if the lender does sue the borrower there is no tax liability since the debt is not forgiven.

The specific tax treatment accorded to the forgiven debt varies by state and depends upon whether the state permits deficiency judgments. In a State that allows deficiency judgments, the amount of the deficiency is counted as ordinary income and should be reported to the IRS as other income on Form 1040. If the homeowner lives in an anti-deficiency State, then there is potentially a capital gain (or loss). In this case, the final sale price of the property is taken to be the total outstanding indebtedness of the homeowner at the time of foreclosure (outstanding mortgage plus uncollected interest and penalties) while the initial sale price is taken to be the initial purchase price adjusted by the amount of the deficiency.⁶⁶

Worked Example

The tax rules associated with the discharge of mortgage indebtedness are best digested through an example. Figure G.1 summarizes the assumptions behind our example.

⁶⁴ For the time being, the tax treatment of mortgage defaults has been overridden by the Mortgage Forgiveness Debt Relief Act of 2007. The Act provides a safe harbor from the taxation of income realized from debt writedowns (through loan modifications or foreclosure sales) for all debt forgiven in 2007, 2008, and 2009. The Act only applies to debt used to build, buy, or improve the principal residence or to refinance debt incurred for these purposes.

⁶⁵ A brief primer on deficiency judgments can be found in Appendix D.

⁶⁶ Generally speaking, the concepts and calculations used in determining tax liabilities for the foreclosure situation also apply to a pre-foreclosure sale or deed-in-lieu of foreclosure.

Figure G.1: Assumptions for Foreclosure Tax Calculation

	<u>Amount</u>
Initial Purchase Price (less transaction costs)	\$200,000
Foreclosure Sale Price	\$150,000
Outstanding Mortgage Debt at Default	\$175,000
Foreclosure Costs	\$9,000
Uncollected Interest/Penalties	\$5,000

Source: Banc of America Securities LLC.

States That Allow Deficiency Judgments

The homeowner must report a home sale at \$150,000 less the lender's foreclosure costs of \$9,000. So, the capital loss to the homeowner is:

$$\begin{aligned}\text{Capital Loss} &= \text{Initial Purchase Price} - (\text{Foreclosure Sale Price} - \text{Foreclosure Costs}) \\ &= \$200,000 - (\$150,000 - \$9,000) = \$59,000\end{aligned}$$

The deficiency amount for the lender is:

$$\begin{aligned}\text{Deficiency} &= \text{Outstanding Mortgage Debt} - (\text{Foreclosure Sale Price} - \text{Foreclosure Costs}) \\ &= \$175,000 - (\$150,000 - \$9,000) = \$34,000\end{aligned}$$

Thus, if the lender chooses not to pursue a deficiency judgment, then the homeowner has to report regular income of \$39,000 (Amount of Deficiency + Uncollected Interest/Penalties) on their tax return (along with a capital loss on the sale of the property).⁶⁷

States That Do Not Allow Deficiency Judgments

In this case, the homeowner reports either a capital gain or loss based on the following calculations:

$$\begin{aligned}\text{Final Sale Price} &= \text{Outstanding Mortgage Debt} + \text{Uncollected Interest/Penalties} \\ &= \$175,000 + \$5,000 = \$180,000\end{aligned}$$

$$\begin{aligned}\text{Initial Basis-for-Sale} &= \text{Initial Purchase Price} - \text{Deficiency Amount} \\ &= \$200,000 - \$34,000 = \$166,000\end{aligned}$$

$$\begin{aligned}\text{Capital Gain (Loss)} &= \text{Final Sale Price} - \text{Initial Basis-for-Sale} \\ &= \$180,000 - \$166,000 = \$14,000\end{aligned}$$

Practical Considerations

There are important qualifications to the tax rule: Borrowers may not have to pay these taxes if they have declared bankruptcy or are insolvent (total debts exceed total assets) at the time of the foreclosure sale.⁶⁸ In practice, many mortgagors may meet this litmus test by the time they

⁶⁷ The IRS only allows investors to claim a write-off for the loss. A loss on a personal residence is considered a nondeductible personal expense for federal income tax purposes.

⁶⁸ Or if the cancellation is intended as a gift.

get to the stage of defaulting on their mortgage. Also, note that the capital gain reported in the anti-deficiency case may not necessarily result in a tax event for the homeowner since capital gains on home sales are shielded for up to \$250,000 (\$500,000 for a married couple filing jointly).⁶⁹

Even if a homeowner does have to pay this tax, the existing rules are not clear as to when it is due. As mentioned earlier, the creditor of a forgiven debt has to file a 1099-C with the IRS and the mortgagor may not have to pay until this is filed. Currently, the IRS does stipulate that the tax to be paid in the year the foreclosure was filed but some tax experts have taken the position that the homeowner is not obligated to pay taxes until they have been officially informed by the lender that the debt has been forgiven and no deficiency judgment will be pursued. Otherwise, the borrower may potentially face the double whammy of paying the tax and being sued by the lender.

⁶⁹ Prior to 1997, several rules permitted avoidance or deferral of the tax on the gain on owner-occupied home sales including the ability to defer gain until a subsequent house was sold (rollover treatment) and provision allowing a one-time exclusion of \$125,000 for those 55 and over. These provisions were replaced with a general \$500,000 exclusion for a married couple filing jointly (\$250,000 for a single individual). The property has to be the primary residence of the owner for two of the five years before the date of sale. The two years of residence do not need to be continuous. Also, the ownership and use tests can be satisfied during different two-year periods. However, you must meet both tests during the 5-year period ending on the date of the sale. There are allowances and exceptions for military service, disability, partial residence and other reasons.

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